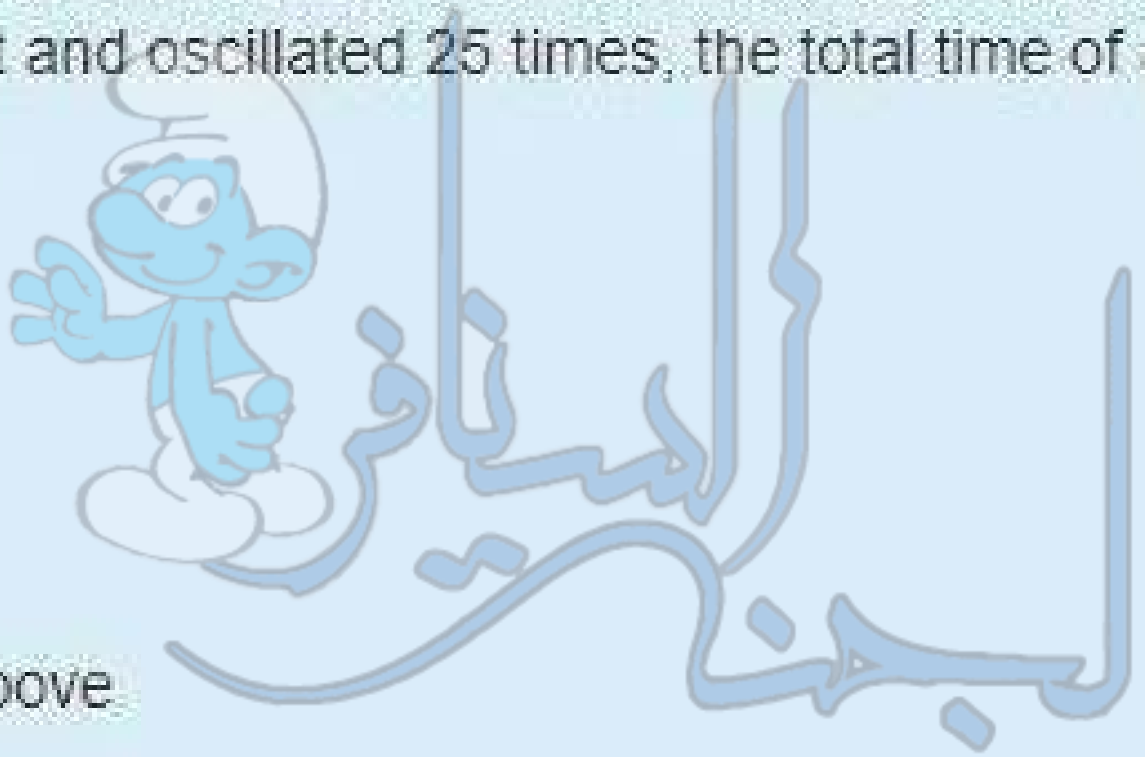
In the simple harmonic oscillator, a mass of (40g) is hanged in a spring, if the extension of the spring is (4 cm) and if the mass drawn to (3cm) from equilibrium point and oscillated 25 times, the total time of an oscillation of 20 times is (in sec):

Select one:

- a. 9.93
- b. 11.91
- c. Non of above
- d. 5.96
- e. 7.95 ✓



In a vector experiment, the ring being large or small does not affect the equilibrium



إختر واحداً:

- صح
- خطأ

Two forces are in equilibrium, if you know that (F_{1X} = 0.25 N), (F_{1Y} = - 0.4 N) then the magnitude of (F_2), and its deviation from the positive (X) axis

Select one:

- a. (0.47 N, 57.99°)
- b. (0.47 N, 237.99°)
- c. (0.47 N, 302.01°)
- d. (0.47 N, 122.01°)

Two forces are in equilibrium, if you know that (F_{1x} = 0.25 N), (F_{1Y} = - 0.4 N) then the magnitude of (F_2), and its deviation from the positive (X) axis



إختر أحد الخيارات:

a. (0.47 N, 57.99°)

b. (0.47 N, 237.99°)

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Select one:

- a. 9.93
- b. 11.91
- c. Non of above
- d. 5.96
- e. 7.95 ✓



The correct answer is: 7.95

In uniform circular motion, the peed is constant but the velocity is variable

Select one:

- a. True ✓
- b. False



The correct answer is: True

In spring constant experiment, if the hanging mass 80g and the elongation in the spring is 4cm, the spring constant (N/m) is:

Select one:

- a. 40
- b. 20 ✓
- c. 10
- d. Non of above
- e. 5



The correct answer is: 20

In vector experiment, the mass hangers do not affect on the equilibrium state

Select one:

- a. False ✓
- b. True



The correct answer is: False

We can conclude from the frictional force that μ_k is smaller than μ_s

Select one:

- a. True ✓
- b. False



The correct answer is: True

The micrometer is more accurate than the vernier caliper

Select one:

- a. False
- b. True ✓



The correct answer is: True

The using of the air track in Newtons second law experiment is to decrease the friction effects

Select one:

- a. True ✓
- b. False



The correct answer is: True

In the elastic collisions, the linear momentum is conserved, while the kinetic energy is not:

Select one:

- a. True
- b. False ✓



The correct answer is: False

The best equipment to measure the thickness of the paper is the Vernier caliper

Select one:

- a. True
- b. False ✓



The correct answer is: False

The sum of the change in kinetic and potential energy is always positive

Select one:

- a. True
- b. False ✓

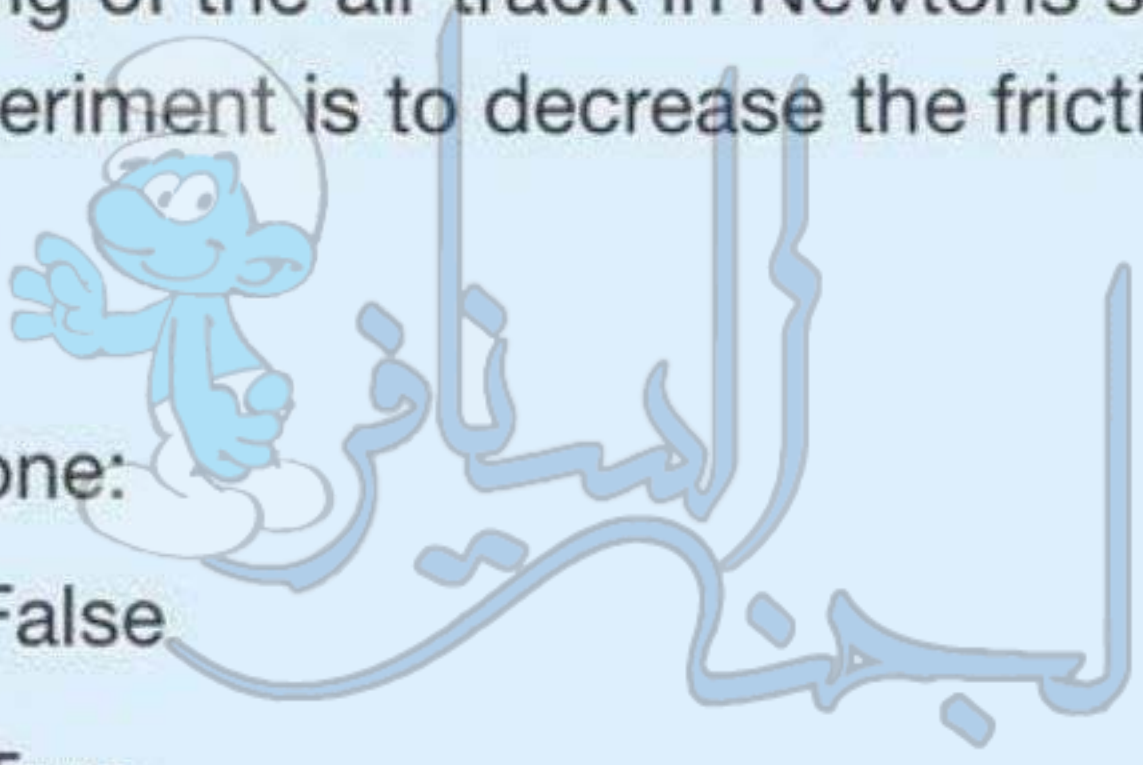


The correct answer is: False

The using of the air track in Newtons second law experiment is to decrease the friction effects

Select one:

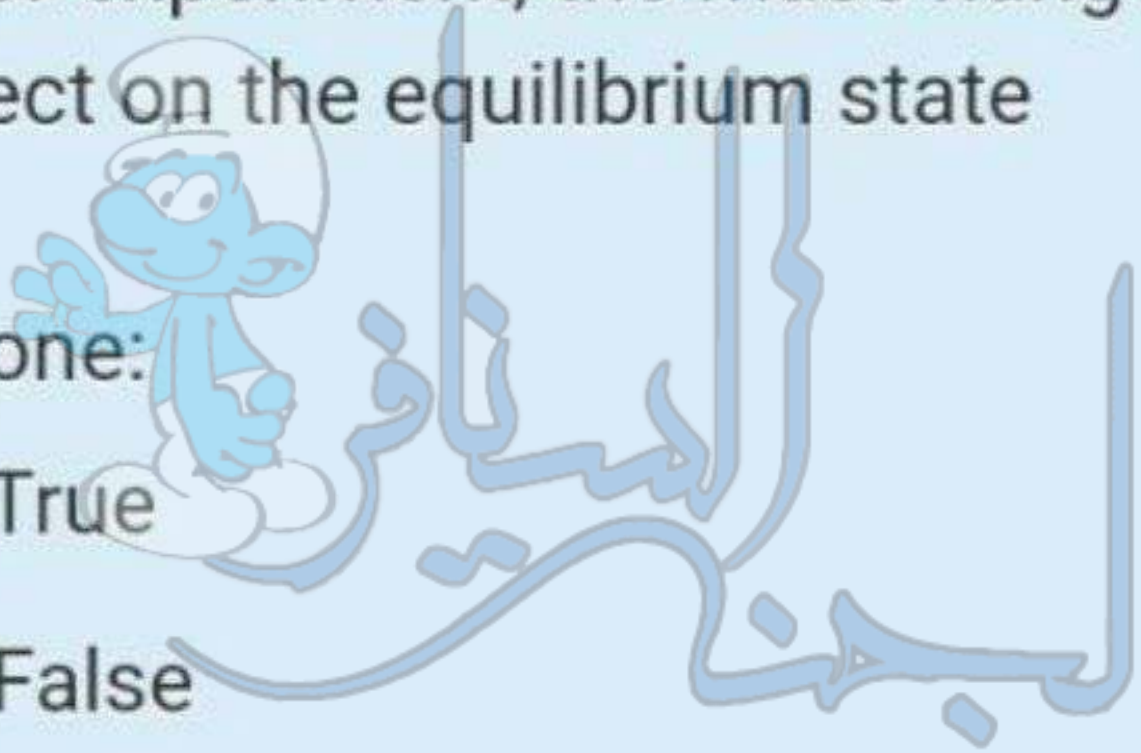
- a. False
- b. True



In vector experiment, the mass hangers do not affect on the equilibrium state

Select one:

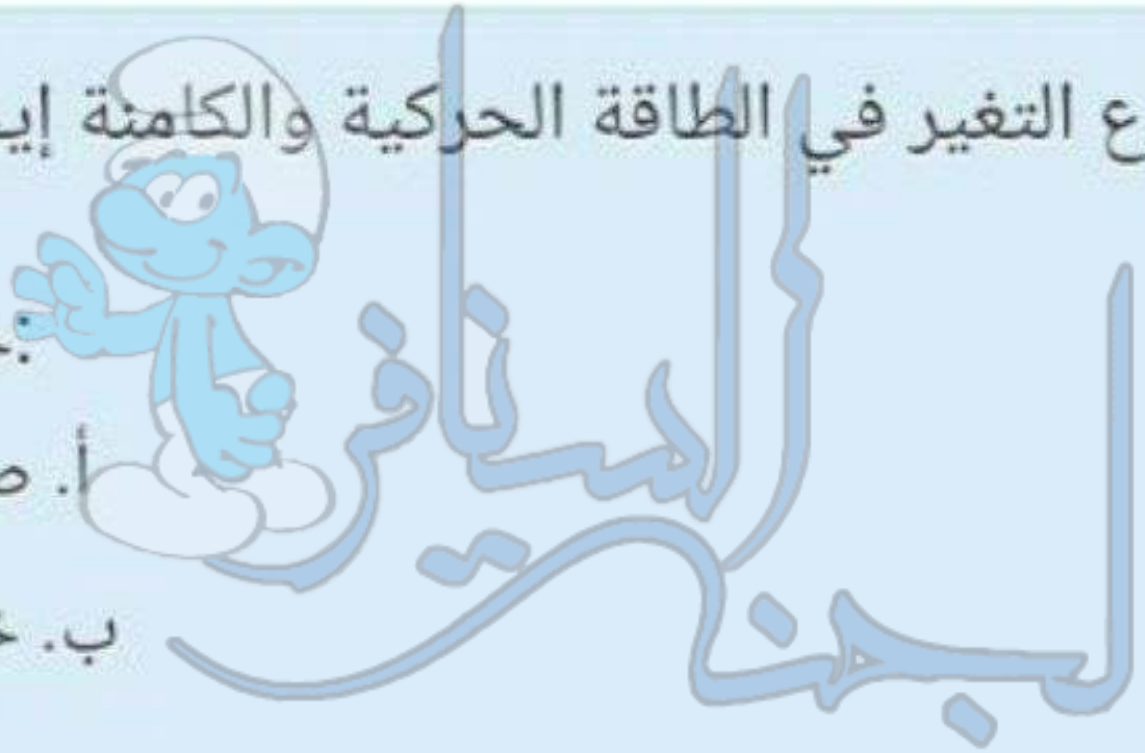
- a. True
- b. False



مجموع التغير في الطاقة الحركية والكامنة إيجابي دائمًا

حدد واحدًا:

- أ. صحيح
- ب. خاطئة



In uniform circular motion, the peed is constant but the velocity is variable

Select one:

- a. False
- b. True

It is wrong to say my weight 70 kg



اخترأحد الخيارات

- صح
- خطأ

The relationship between the length of the thread of the pendulum and the square of the periodic time

Select one:

- A. linear direct proportional
- B. curve relationship
- C. inverse linear proportional
- D. no relation

In a gravitational potential energy experiment, when (m) reaches Earth, the mass (M)

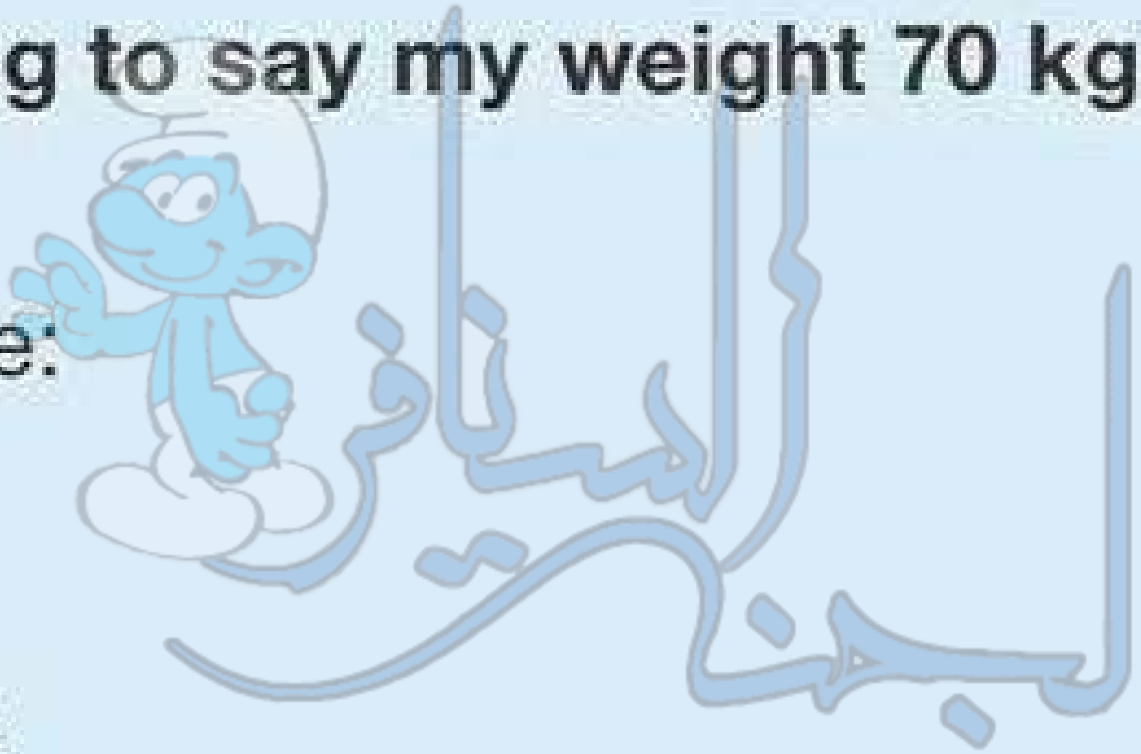
Select one:

- a. **The mass(M) will stop.**
- b. **The speed of the mass (M) will increase.**
- c. **The speed of the mass (M) will decrease**
- d. **The mass (M) will move in costant speed.**

It is wrong to say my weight 70 kg

Select one:

- True
- False



In the inelastic collisions experiment,we can say

Select one:

- a. **The linear momentum always conserved ($P_f = P_i$).**
- b. **$V(m_1)_{final} = V(m_1)_{initial}$**
- c. **$(K.E_f) \neq (K.E_I)$**
- d. **All of them**

In the inelastic collisions experiment,we can say

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The most suitable tool for measuring the thickness of a razor blade is vernnier caliper.

اخترأحد الخيارات

- صح
- خطأ

In grvitiational potential energy experiment, always Δ(K.E) = Δ(U.E).



اخترأحد الخيارات

- صح
- خطأ

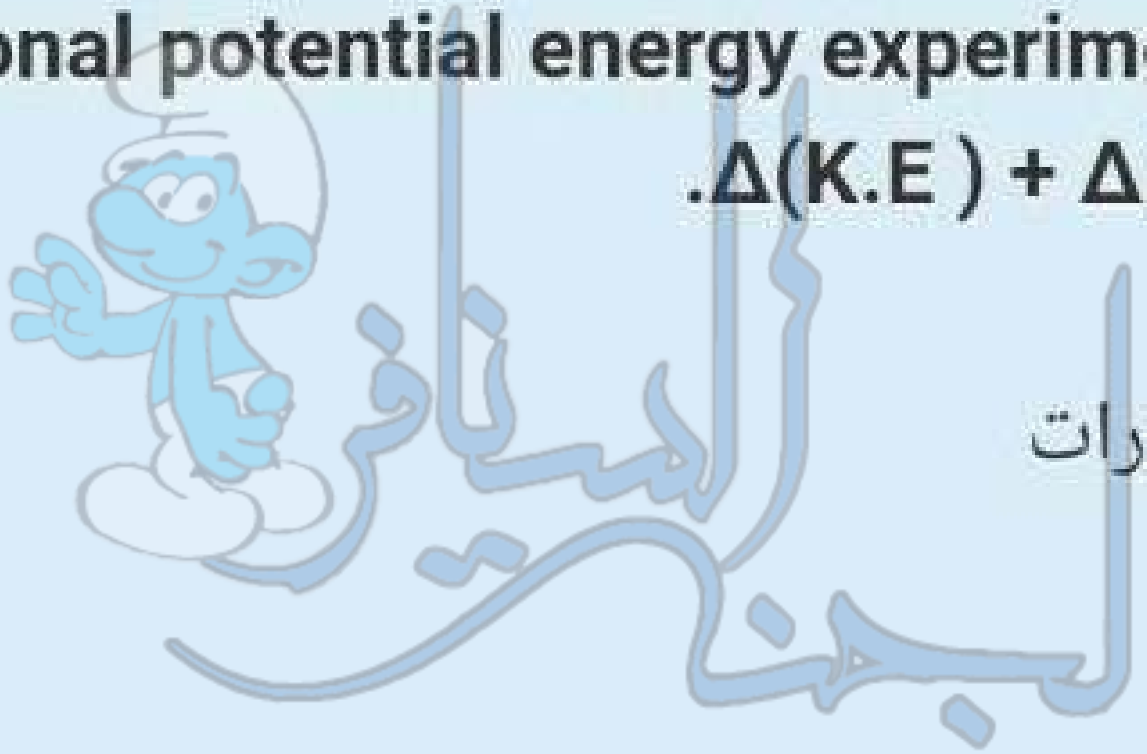
In grvitiational potential energy experiment, always Δ(K.E) = Δ(U.E).



اخترأحد الخيارات

- صح
- خطأ

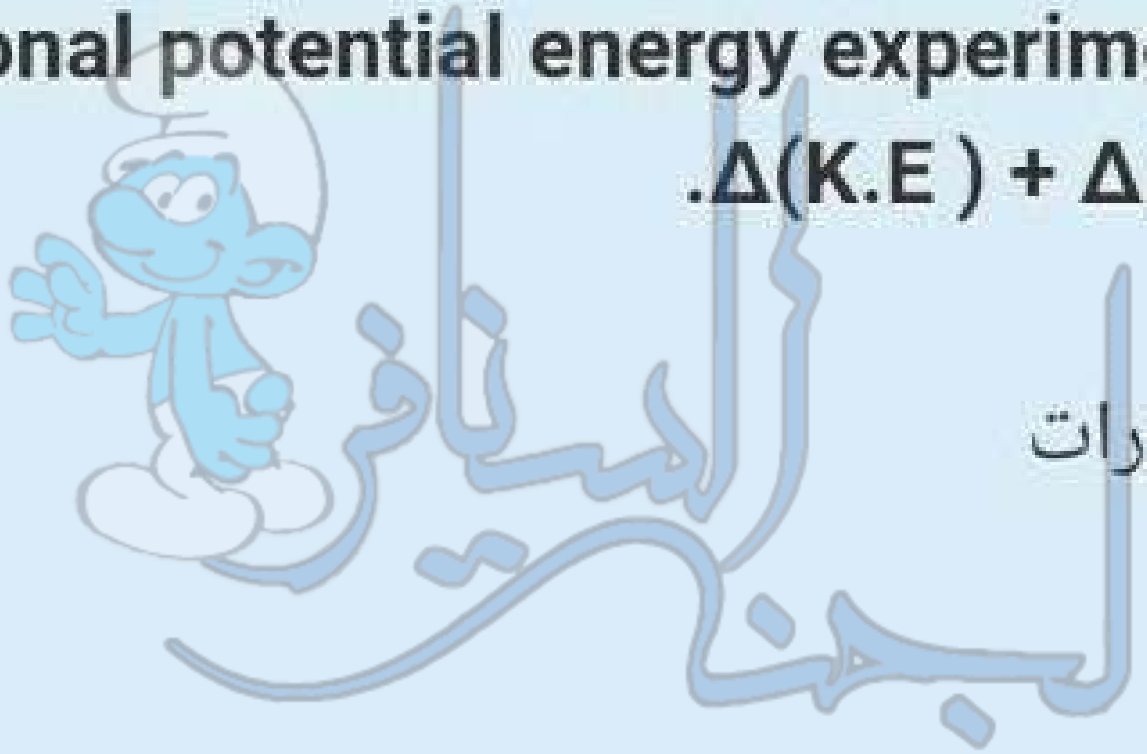
In grvitiational potential energy experiment, always
.Δ(K.E) + Δ(U.E)=zero



اخترأحد الخيارات

- صح
- خطأ

In grvitiational potential energy experiment, always
.Δ(K.E) + Δ(U.E)=zero



اخترأحد الخيارات

- صح
- خطأ

Suspend a body of mass (30 g) with a spring, if you know that the periodic time for the existence of this mass and when the elongation is fixed is 0.37 s.

The spring constant (K) is:

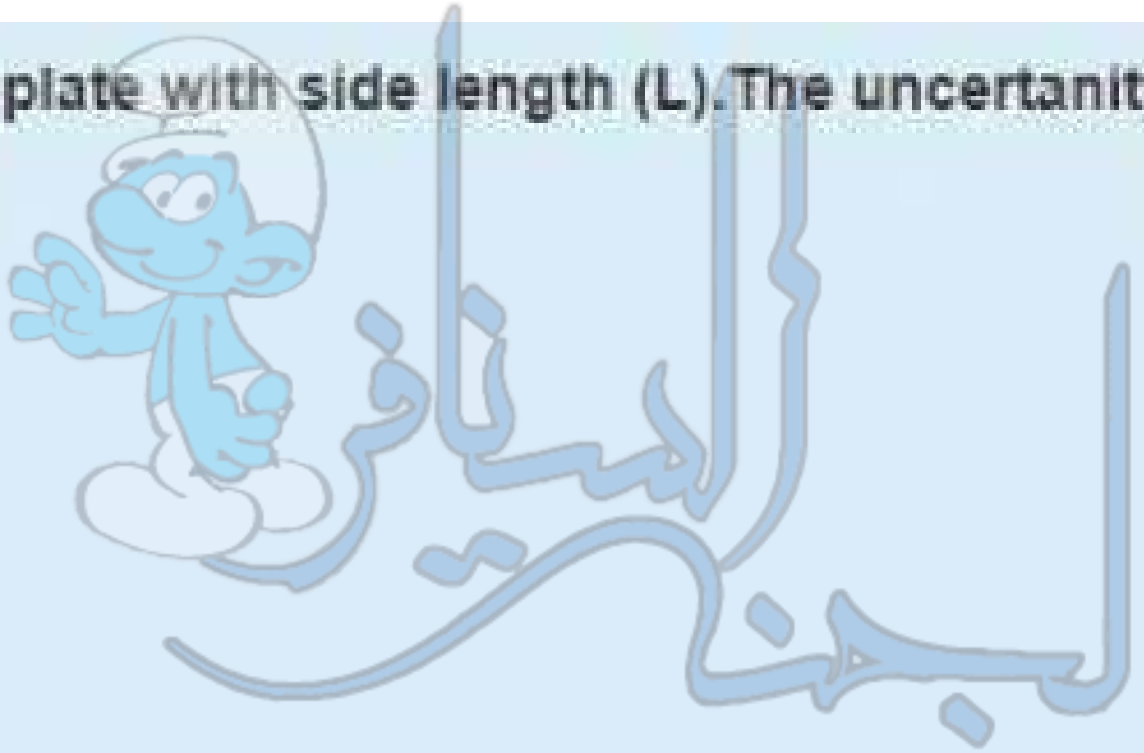
Select one:

- a. **8.6 N/m**
- b. **6 N/m**
- c. **10 N/m**
- d. **12 N/m**

Square-shaped plate with side length (L).The uncertainty (error) in it area is:

Select one:

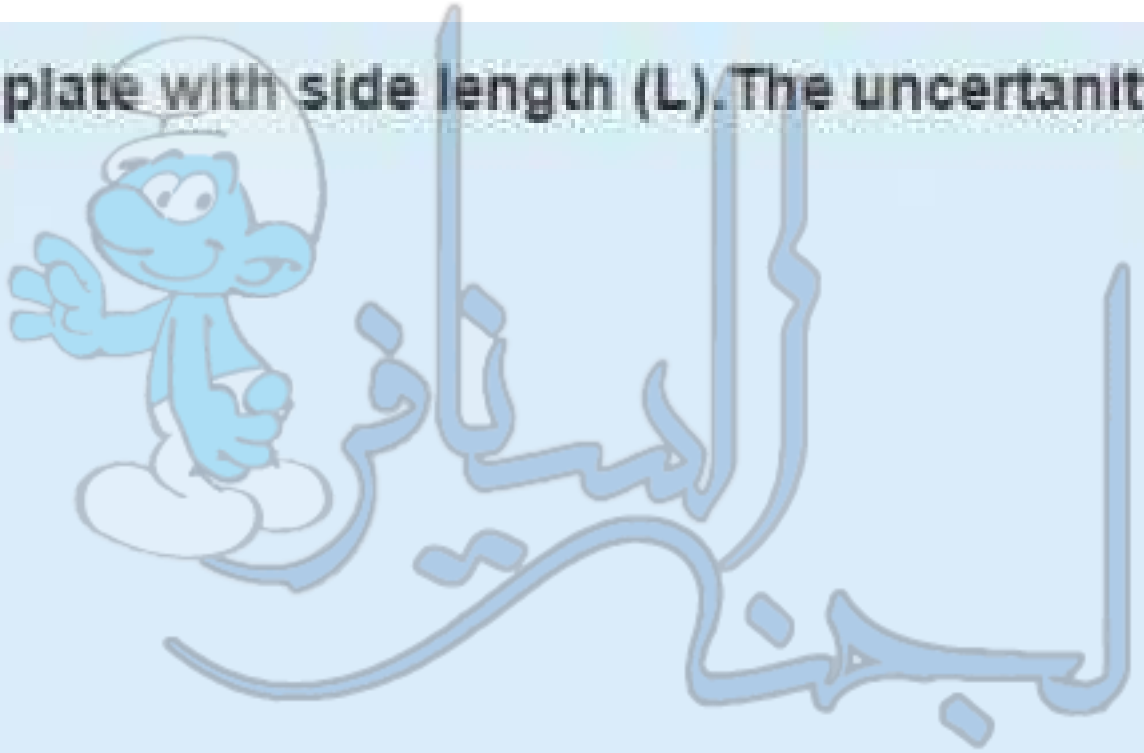
- a. $4L.\Delta L$
- b. $4L^2.\Delta L$
- c. $2L^2.\Delta L$
- d. $2L.\Delta L$



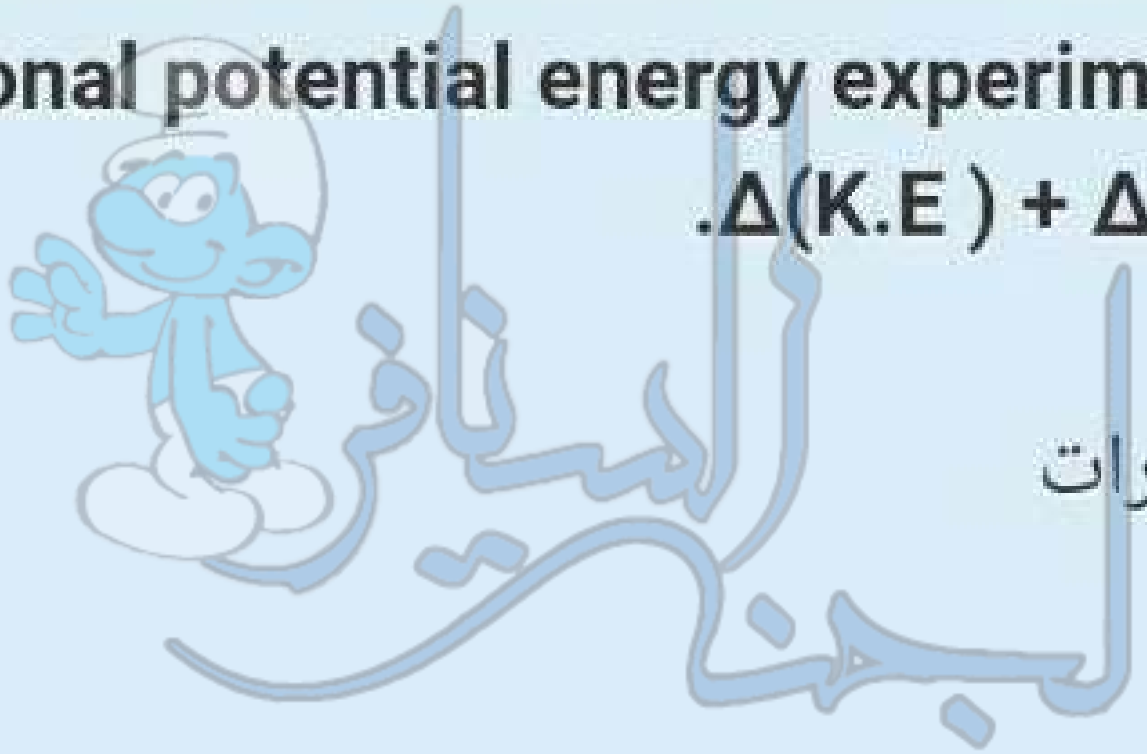
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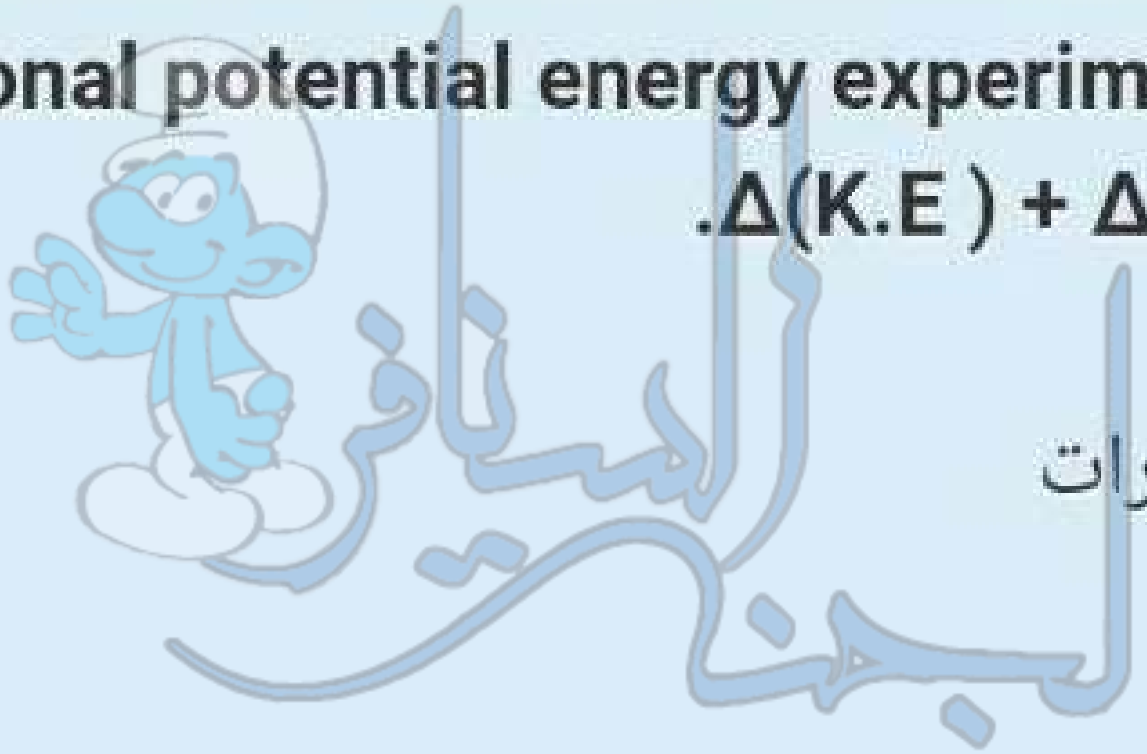
In grvitiational potential energy experiment, always $\Delta(K.E) + \Delta(U.E) = zero$.



اختراحد الخيارات

- صح
- خطأ

In grvitiational potential energy experiment, always $\Delta(K.E) + \Delta(U.E) = zero$.



اختراحد الخيارات

- صح
- خطأ

The most appropriate tool to measure the depth of water in a cup is vernnier caliper.

Select one:

- True
- False

The most appropriate tool to measure the depth of water in a cup is vernnier caliper.

Select one:

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It is wrong to say my weight 70 N.

Select one:

- True
- False

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- True
- False

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Select one:

- True
- False

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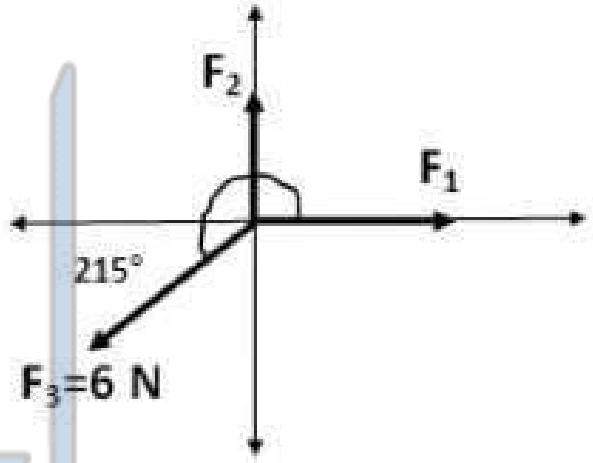
Select one:

- True
- False

Three forces are in equilibrium, depending on the given figure. The value of (F_2) is

Select one:

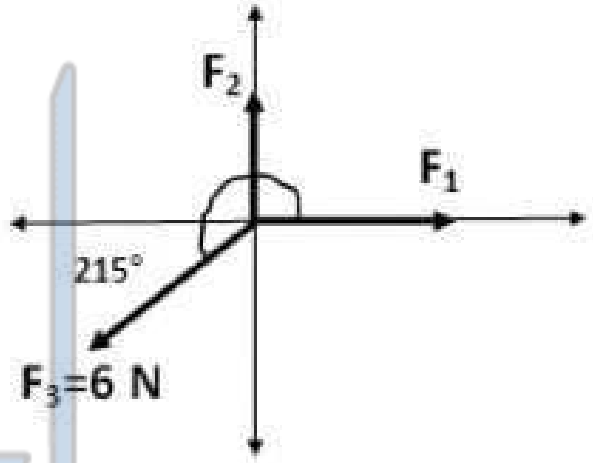
- a. **9.3 N**
- b. **3.44 N**
- c. **8.5 N**
- d. **6 N**



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Select one:

- a. **9.3 N**
- b. **3.44 N**
- c. **8.5 N**
- d. **6 N**



In a gravitational potential energy experiment, (when (m) reaches Earth, the mass (M

اخترأحد الخيارات

a. **The speed of the mass (M) will decrease**

.b. **The mass(M) will stop**

.c. **The mass (M) will move in costant speed**

.d. **The speed of the mass (M) will increase**

In a gravitational potential energy experiment, (when (m) reaches Earth, the mass (M

اخترأحد الخيارات

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.b. **The mass(M) will stop**

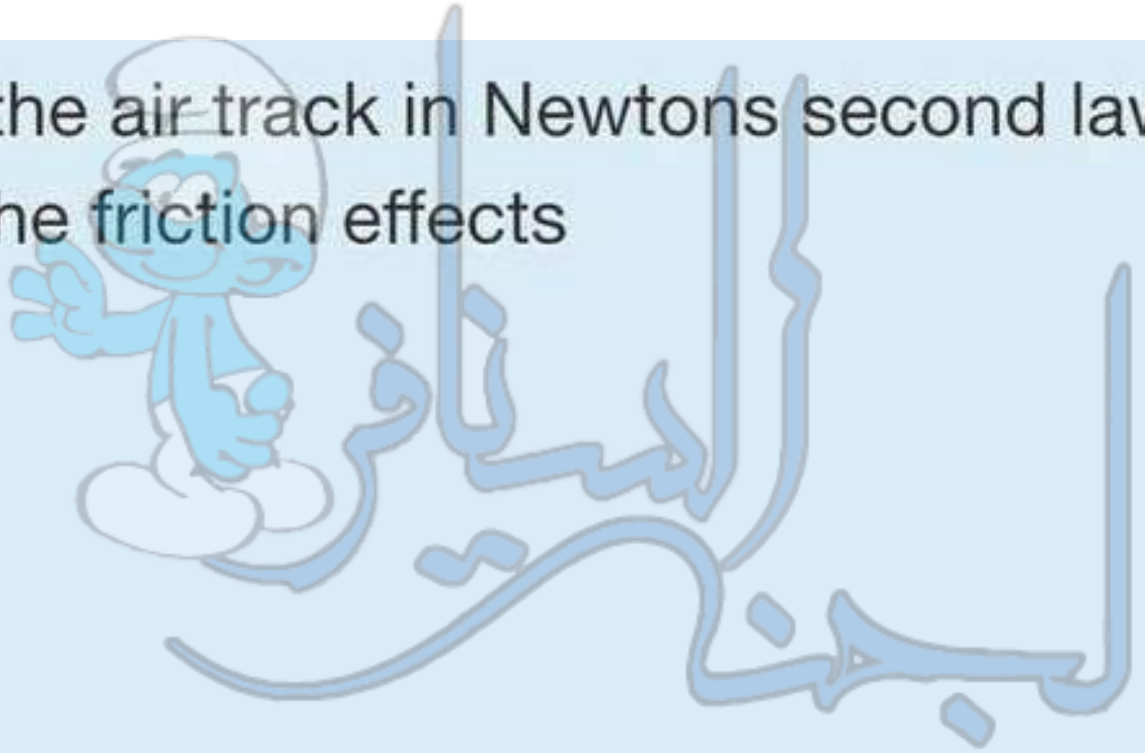
.c. **The mass (M) will move in costant speed**

.d. **The speed of the mass (M) will increase**

The using of the air track in Newtons second law experiment is to decrease the friction effects

Select one:

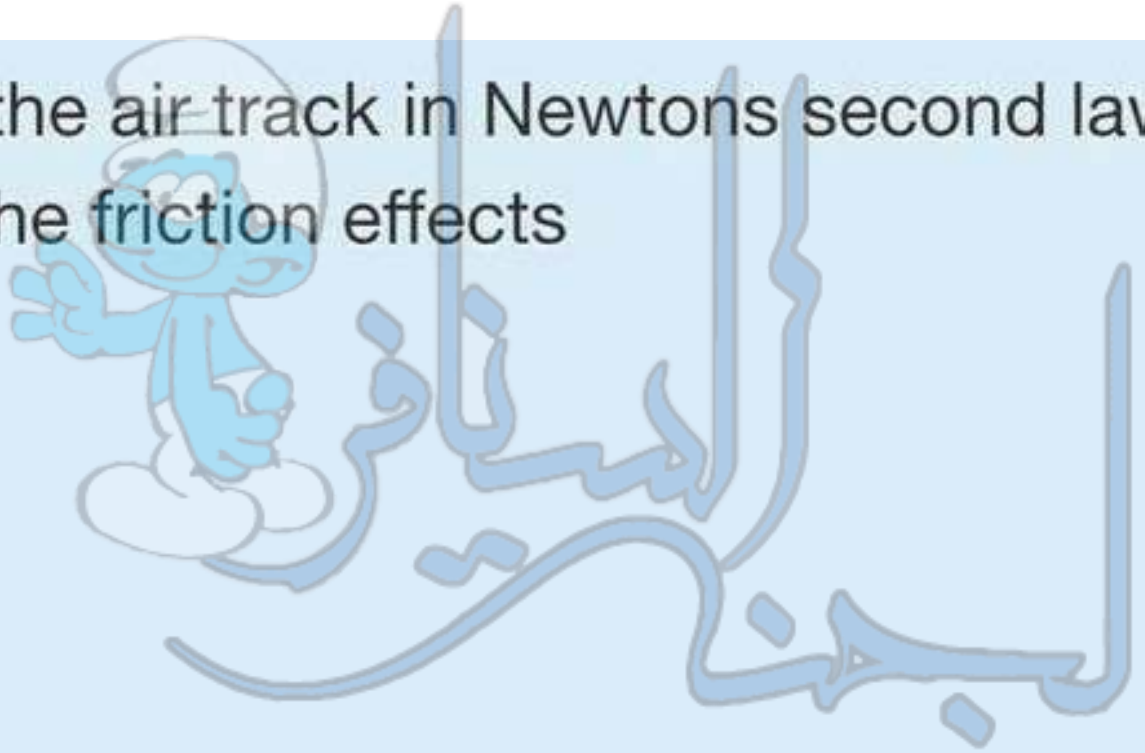
- a. True
- b. False



The using of the air track in Newtons second law experiment is to decrease the friction effects

Select one:

- a. True
- b. False



In grvitiational potential energy experiment,

$(K.E_{f_{(M,m)}}) = (U.E_{i_{(m)}})$.

Select one:

- True
- False



In grvitiational potential energy experiment,

$(K.E_{f_{(M,m)}}) = (U.E_{i_{(m)}})$.

Select one:

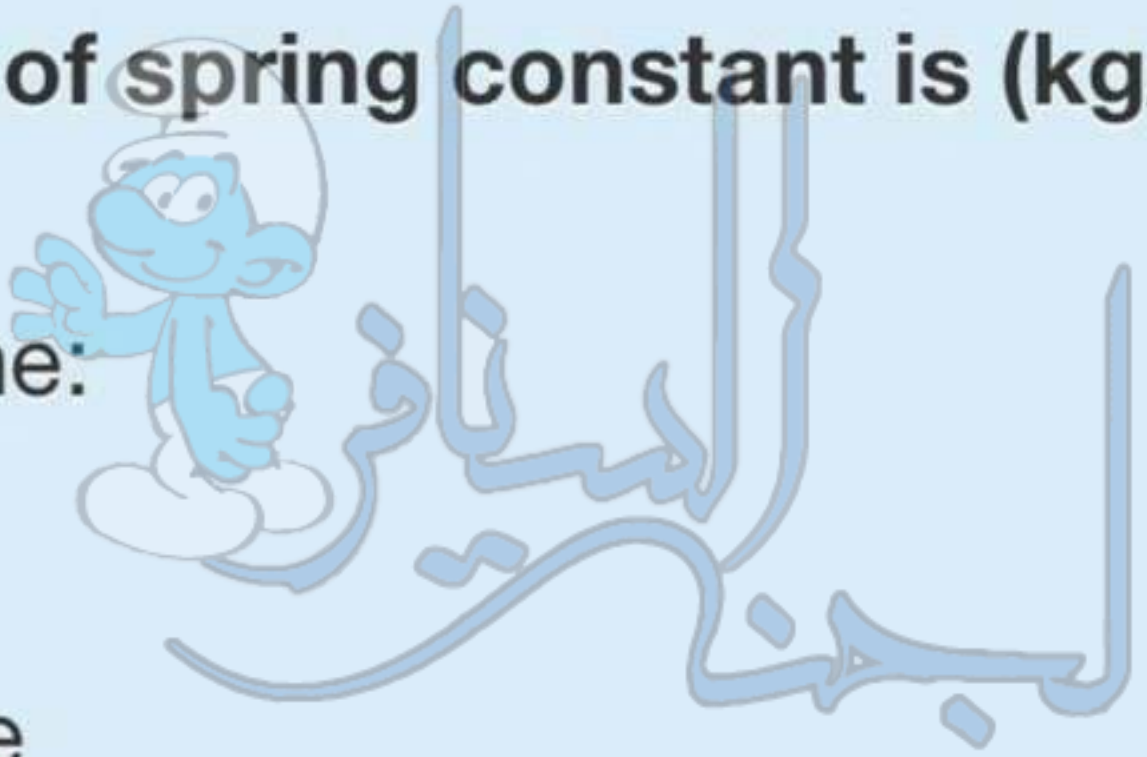
- True
- False



The unit of spring constant is (kg/s^2).

Select one:

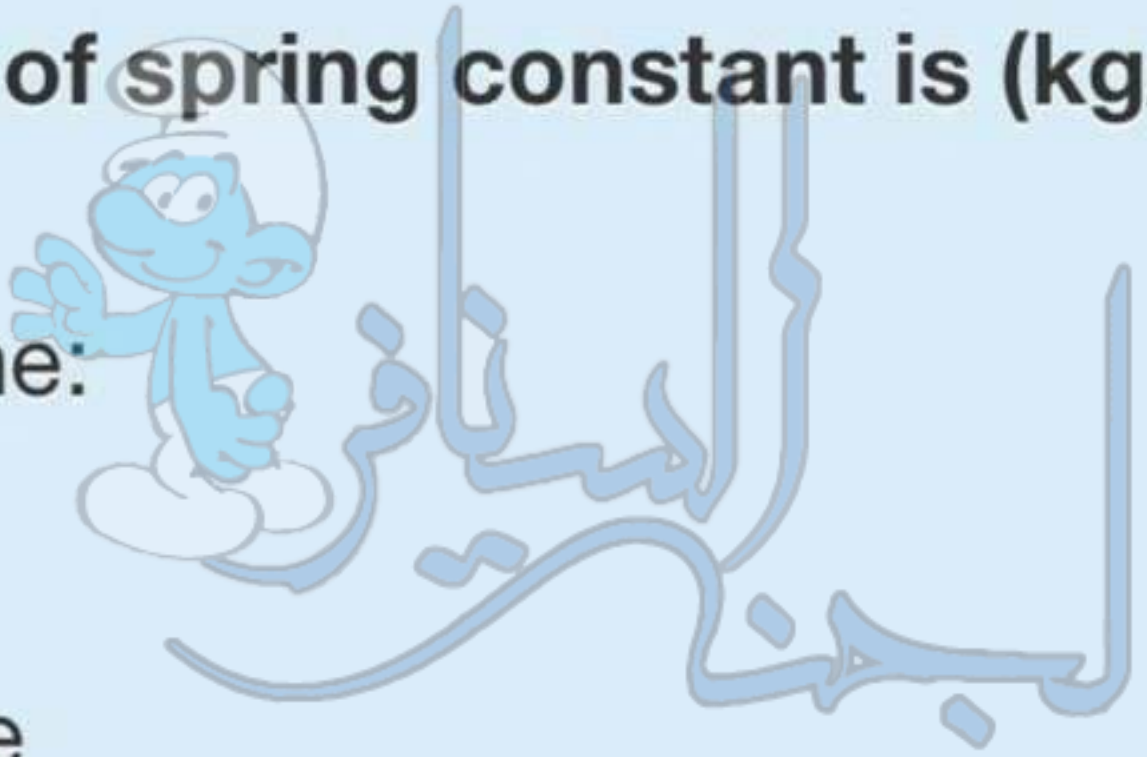
- True
- False



The unit of spring constant is (kg/s^2).

Select one:

- True
- False



The number of significant digits in this number ($0.00280X10^{5}$) is 3 (s.f)

Select one:

- True
- False

The number of significant digits in this number ($0.00280X10^{5}$) is 3 (s.f)

Select one:

- True
- False

: we can say about (μs) and (μk) that

اخترأحد الخيارات

- (A. (μs)< (μk
- (B. (μs) > (μk
- C. -1≥ (μs)AND (μk) ≥ 1
- D. (μs)AND (μk) ≥ 1



: we can say about (μs) and (μk) that

اخترأحد الخيارات

- (A. (μs)< (μk
- (B. (μs) > (μk
- C. -1≥ (μs)AND (μk) ≥ 1
- D. (μs)AND (μk) ≥ 1



A ball mass (m),and raduis (r).The uncertanity (error) of its volume

Select one:

- a. $8\pi.r^{3}.\Delta r$
- b. $8\pi.r.m.\Delta r$
- c. $8\pi.r^{2}.\Delta r$
- d. $4\pi.r^{2}.\Delta r$

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Select one:

- a. $8\pi.r^{3}.\Delta r$
- b. $8\pi.r.m.\Delta r$
- c. $8\pi.r^{2}.\Delta r$
- d. $4\pi.r^{2}.\Delta r$

Suspend a body of mass (30 g) with a spring, if you know that the periodic time for the existence of this mass and when the elongation is fixed is 0.37 s.

The spring constant (K) is:

Select one:

- a. **10 N/m**
- b. **12 N/m**
- c. **6 N/m**
- d. **8.6 N/m**

Suspend a body of mass (30 g) with a spring, if you know that the periodic time for the existence of this mass and when the elongation is fixed is 0.37 s.

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In the inelastic collisions experiment,we can say

اخترأحد الخيارات

a. **The linear momentum always conserved ($P_f = P_i$).**

b. $V(m_1)_{final} = V(m_1)_{initial}$

c. $(K.E_f) \neq (K.E_I)$

d. **All of them**

Using the rules of significant numbers (3.025+2.4=5.4)

Select one:

- True
- False

It is wrong to say my weight 70 N.

Select one:

- True
- False



It is wrong to say my weight 70 N.

Select one:

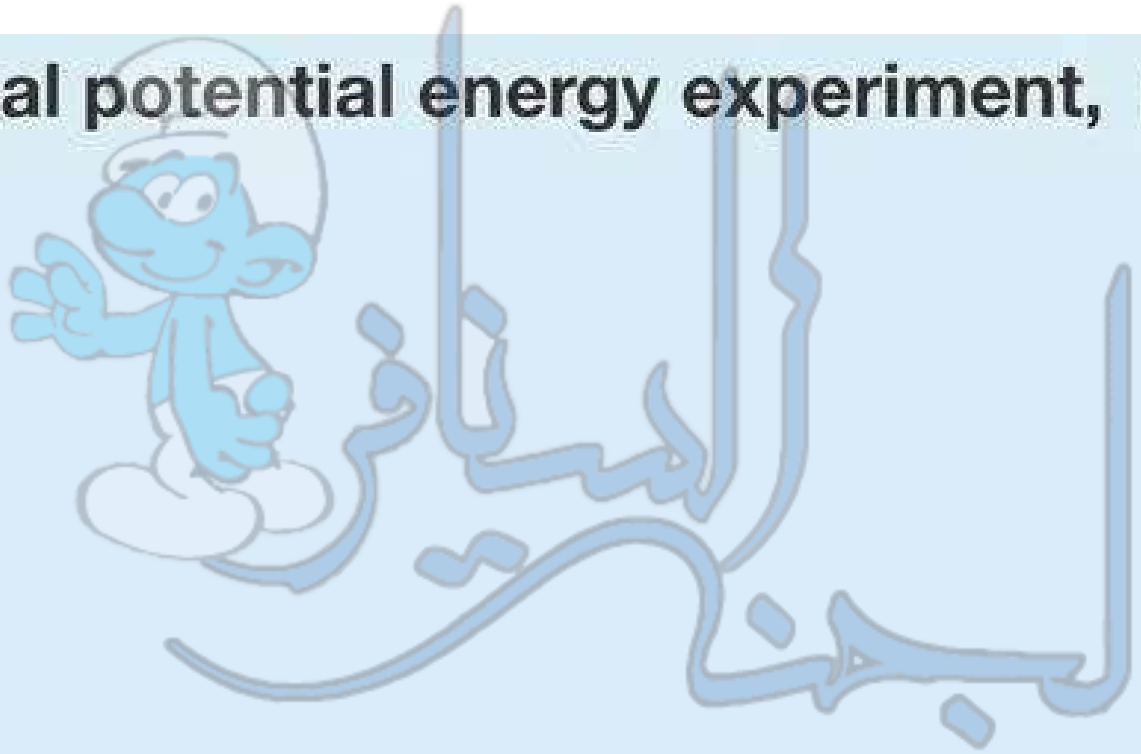
- True
- False



In grvitiational potential energy experiment, $(K.E_{f_{(M,m)}}) = (U.E_{i_{(m)}})$

Select one:

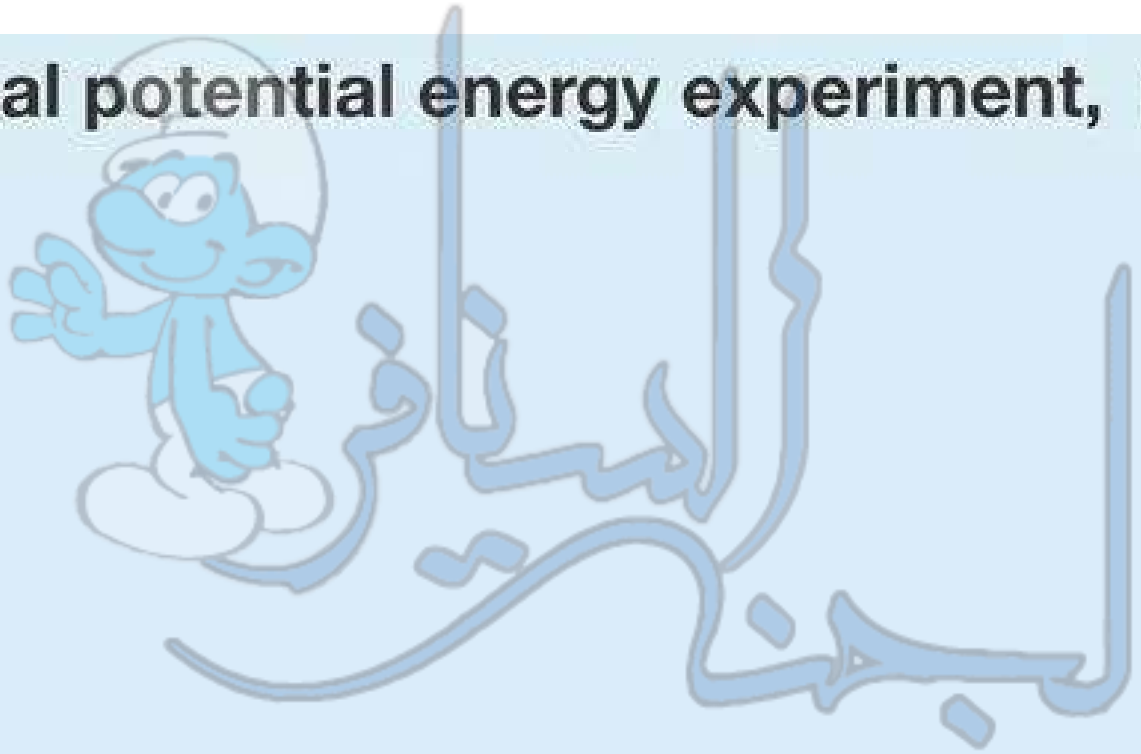
- True
- False



In grvitiational potential energy experiment, $(K.E_{f_{(M,m)}}) = (U.E_{i_{(m)}})$

Select one:

- True
- False



Square-shaped plate with side length (L).The uncertanity (error) in it area is:

Select one:

- a. $4L^2.\Delta L$
- b. $4L.\Delta L$
- c. $2L^2.\Delta L$
- d. $2L.\Delta L$



Square-shaped plate with side length (L).The uncertanity (error) in it area is:

Select one:

- a. $4L^2.\Delta L$
- b. $4L.\Delta L$
- c. $2L^2.\Delta L$
- d. $2L.\Delta L$



Using the rules of significant numbers (3.025 - 2.4=0.6)



اخترأحد الخيارات

- صح
- خطأ

The most suitable tool for measuring the thickness of a razor blade is vernnier caliper.

Select one:

- True
- False

The using of the air track in Newtons second law experiment is to decrease the friction effects



اختر أحد الخيارات

a. False

b. True

The most suitable tool for measuring the thickness of a razor blade is micrometer.

Select one:

- True
- False

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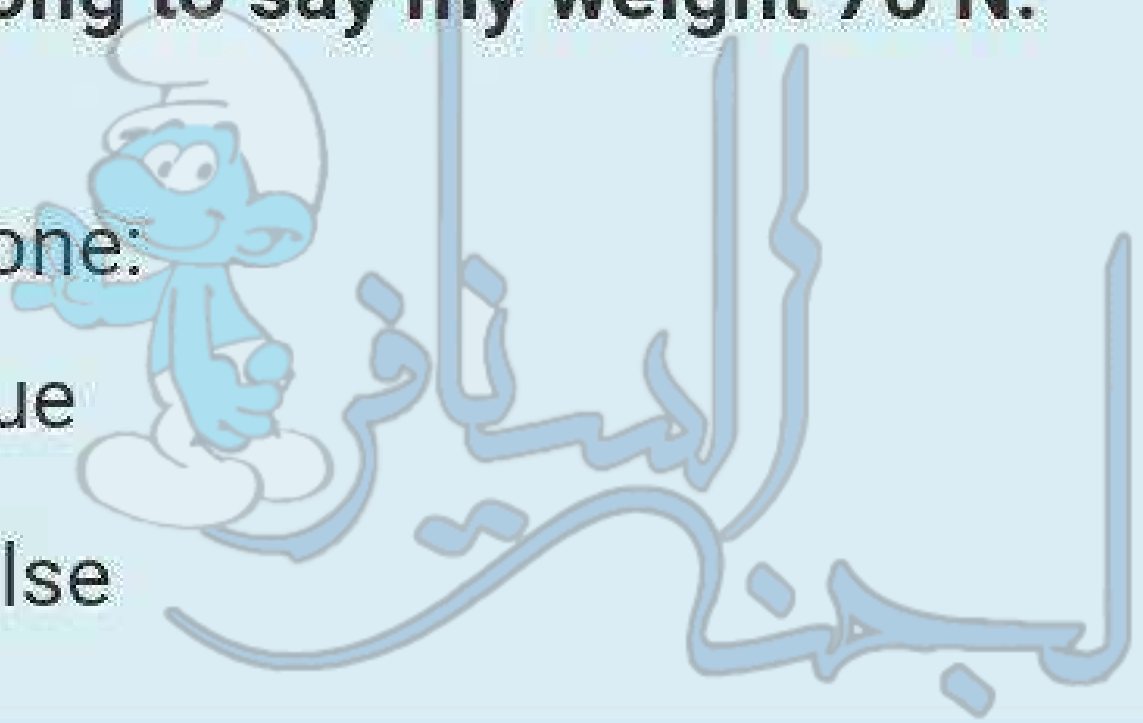
Select one:

- True
- False

It is wrong to say my weight 70 N.

Select one:

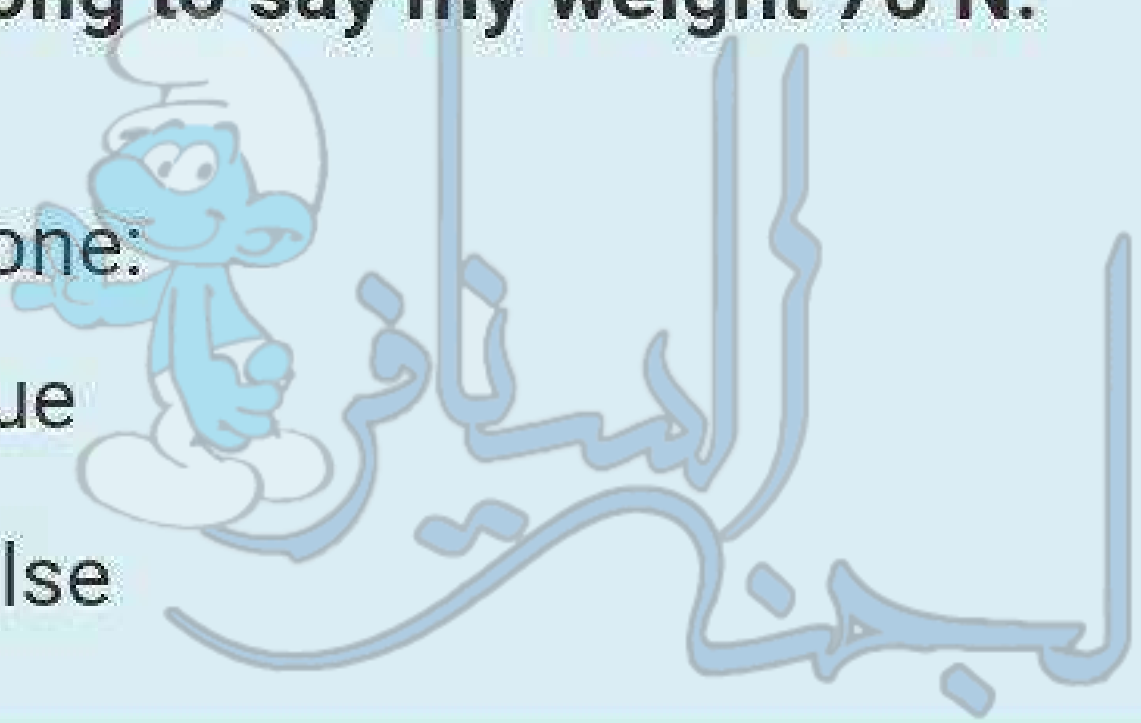
- True
- False



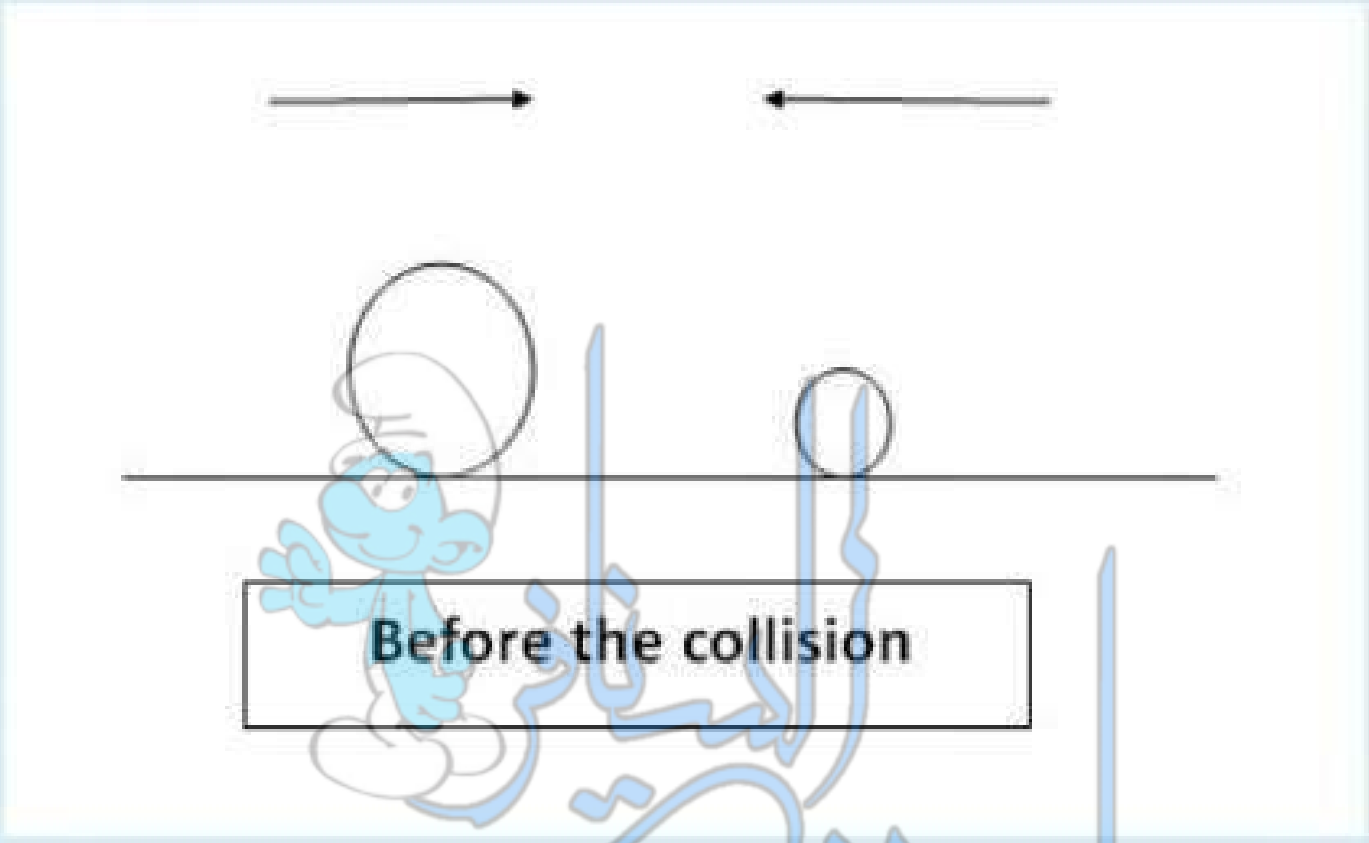
It is wrong to say my weight 70 N.

Select one:

- True
- False



In the elastic collisons,if (M=3m) and $V_{1i\ (M)}$= 2 m/s, $V_{2i(m)}$ = 3 m/s , as the final linear momentum (P_f = 18 kg.m/s) . then (M) is:

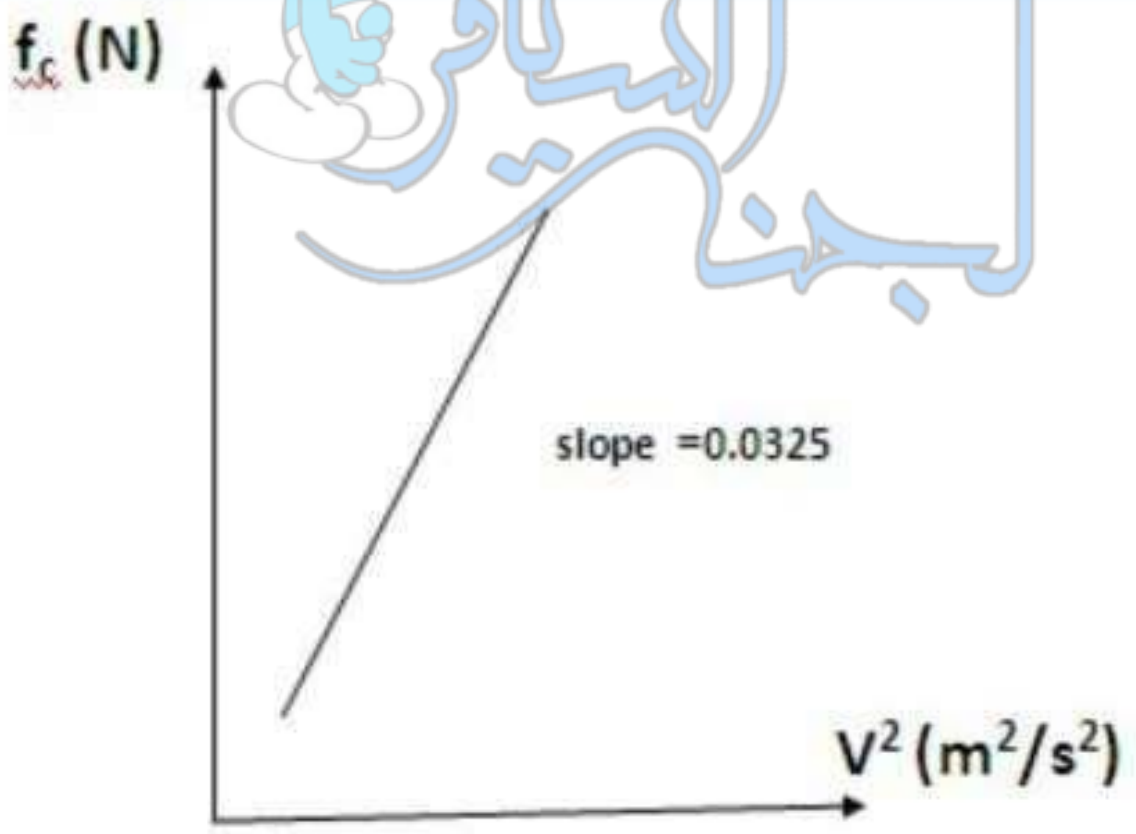


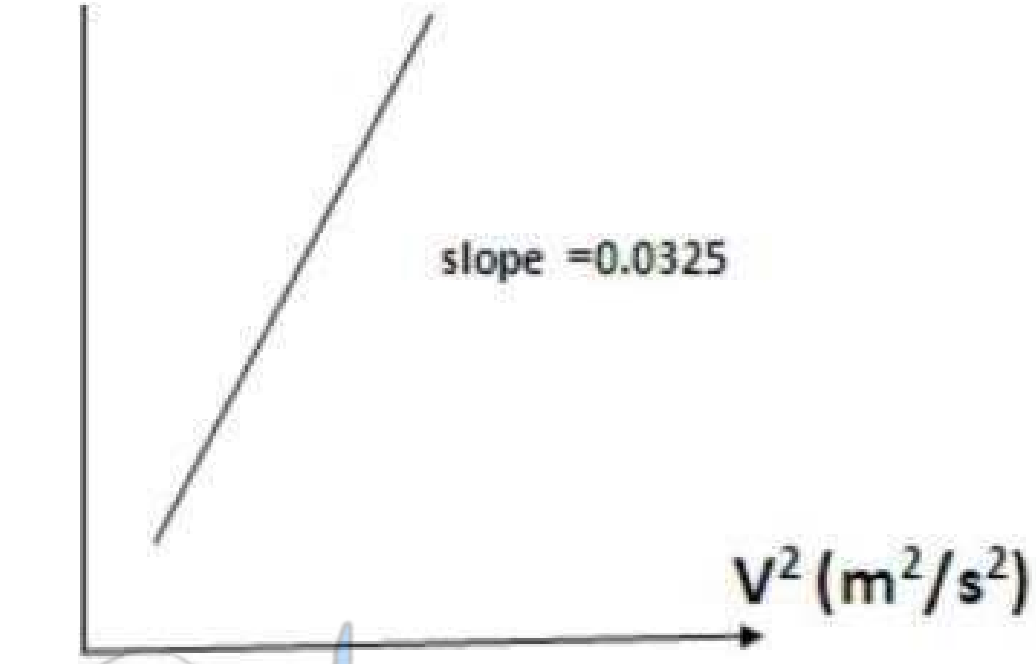
Select one:

- a. 12 kg
- b. 4 kg
- c. 18 kg
- d. 6 kg

In the experiment of uniform circular motion, the student obtained the shown drawing, when the radius was fixed at (40 cm).

Based on the student's drawing, the mass (m), which was constant in the experiment, is equal to:





Select one:

- a. **13 g**
- b. **10 g**
- c. **12 g**
- d. **9 g**

In a gravitational potential energy experiment, when (m) reaches Earth, the mass (M)

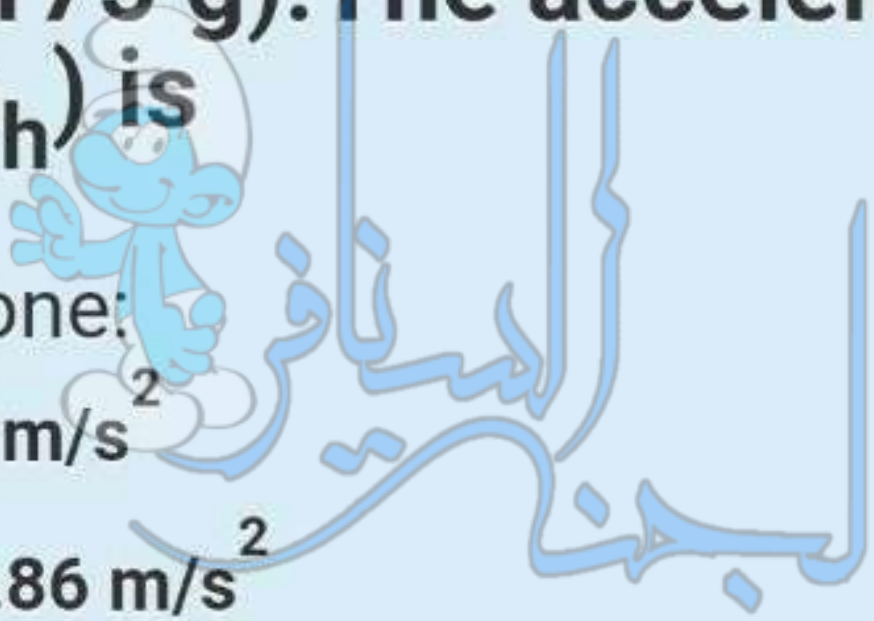
Select one:

- a. **The speed of the mass (M) will increase.**
- b. **The mass(M) will stop.**
- c. **The mass (M) will move in costant speed.**
- d. **The speed of the mass (M) will decrease**

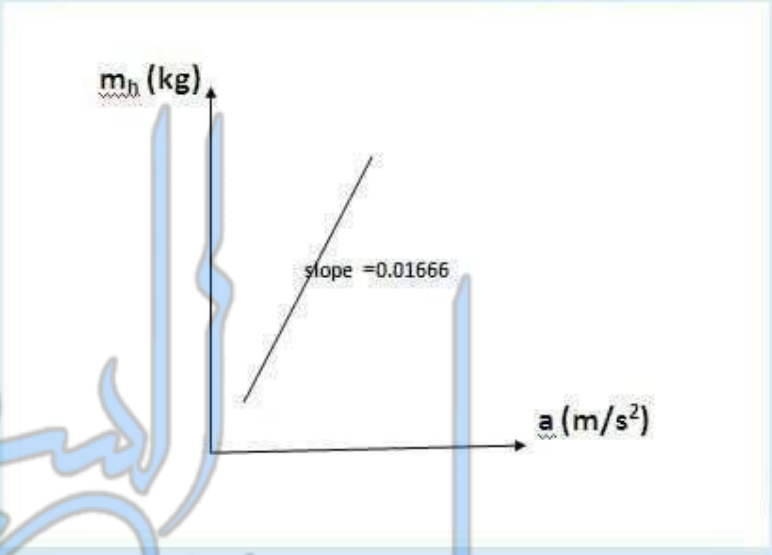
In Newton's second law experiment,when (m_h=15g)and the total mass was (175 g).The acceleration of (m_h) is

Select one:

- a. **2 m/s^2**
- b. **0.86 m/s^2**
- c. **1 m/s^2**
- d. **0.125 m/s^2**



In Newton's second law experiment, a student obtained the graph shown, when the total mass was constant(200g).
Depending on his drawing, the magnitude of the acceleration of the Earth's gravitational pull



اختر أحد الخيارات

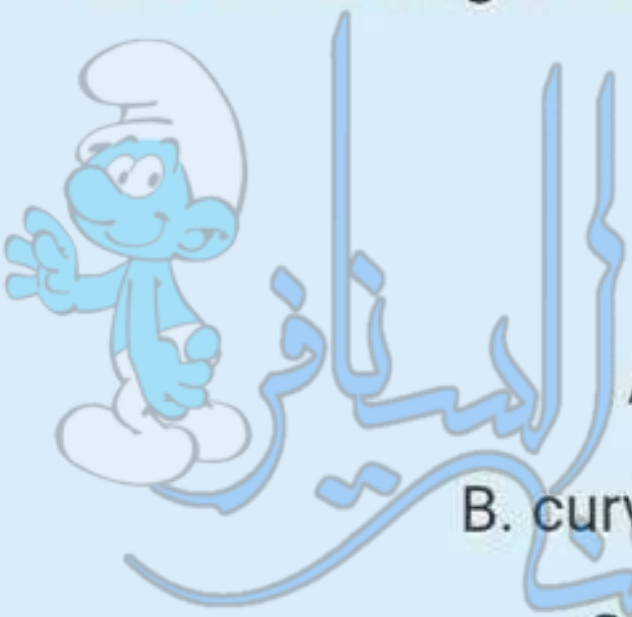
a. 11 m/s^2

b. 9 m/s^2

c. 8 m/s^2

d. 12 m/s^2

The relation between the mass of pendulum (m), and the periodic time (one osiliasion time) (T) at the same lenght of pendulum thread



اخترأحد الخيارات

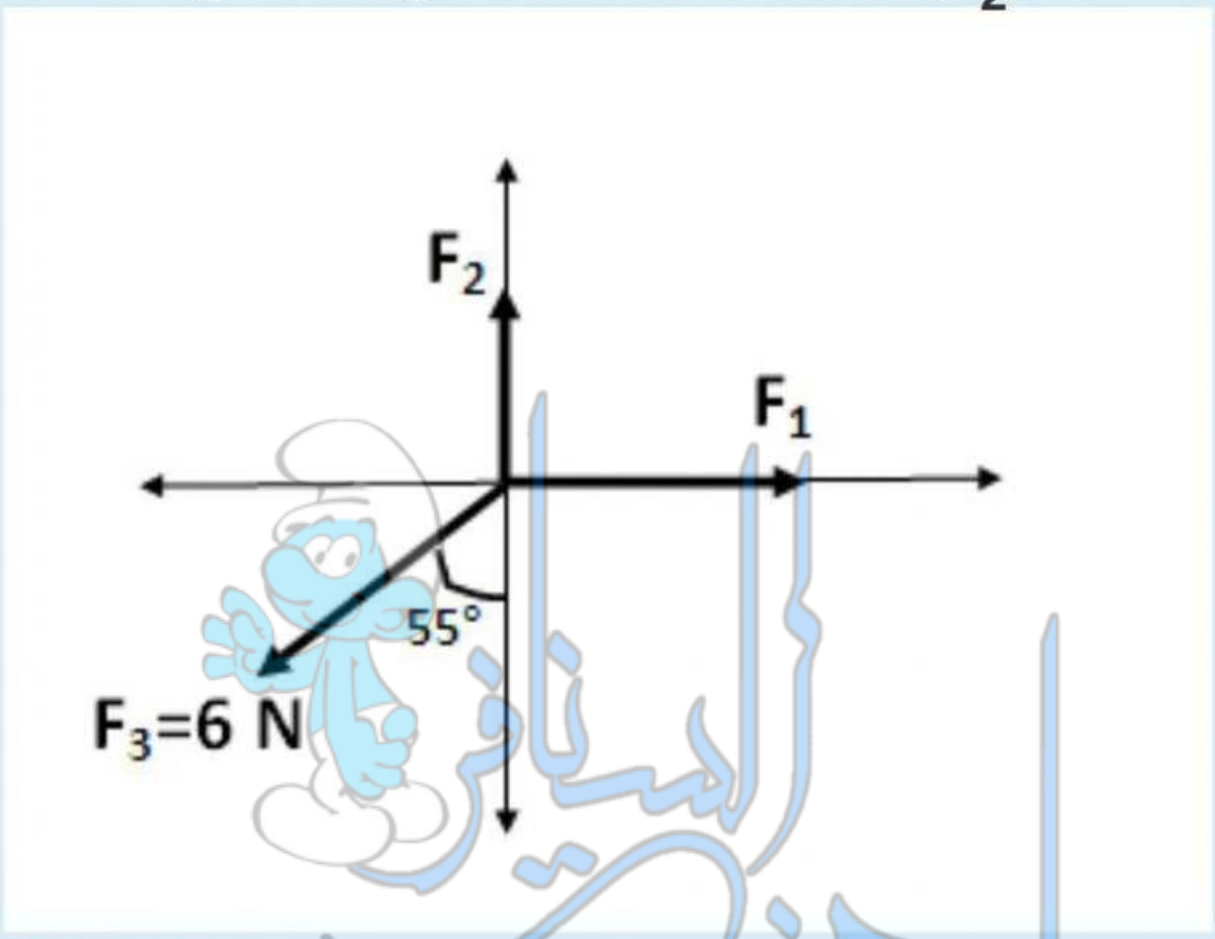
A. linear direct

B. curve relationship

C. inverse linear

D. no relation

Three forces are in equilibrium, depending on the given figure. The value of (F_2) is



Select one:

- a. **8.5 N**
- b. **3.44 N**
- c. **6 N**
- d. **9.3 N**

In a gravitational potential energy experiment, when (m) reaches Earth, the mass (M)

Select one:

- a. **The speed of the mass (M) will decrease**
- b. **The mass (M) will move in costant speed.**
- c. **The mass(M) will stop.**
- d. **The speed of the mass (M) will increase.**

Square-shaped plate with side length (L).The uncertanity (error) in it area is:

Select one:

- a. $4L.\Delta L$
- b. $2L.\Delta L$
- c. $4L^2.\Delta L$
- d. $2L^2.\Delta L$

In Uniform circular motion experiment,the benefit of using

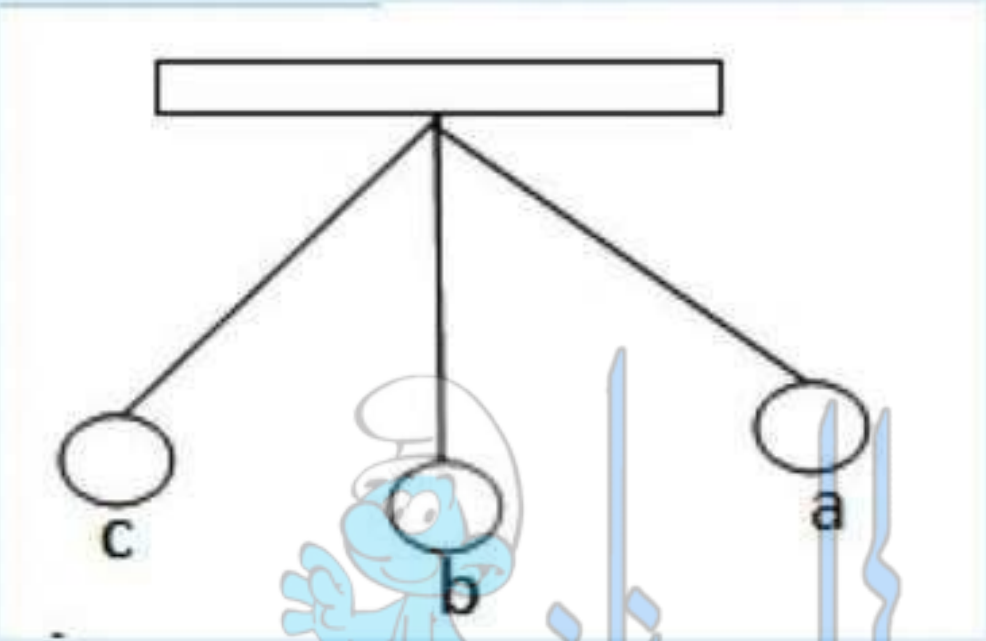
crocodile tongs,is



Select one:

- a. **To keep ($Mg=f_c$).**
- b. **To keep the raduis constant.**
- c. **To keep the magnitued of the velocity constant.**
- d. **All of them**

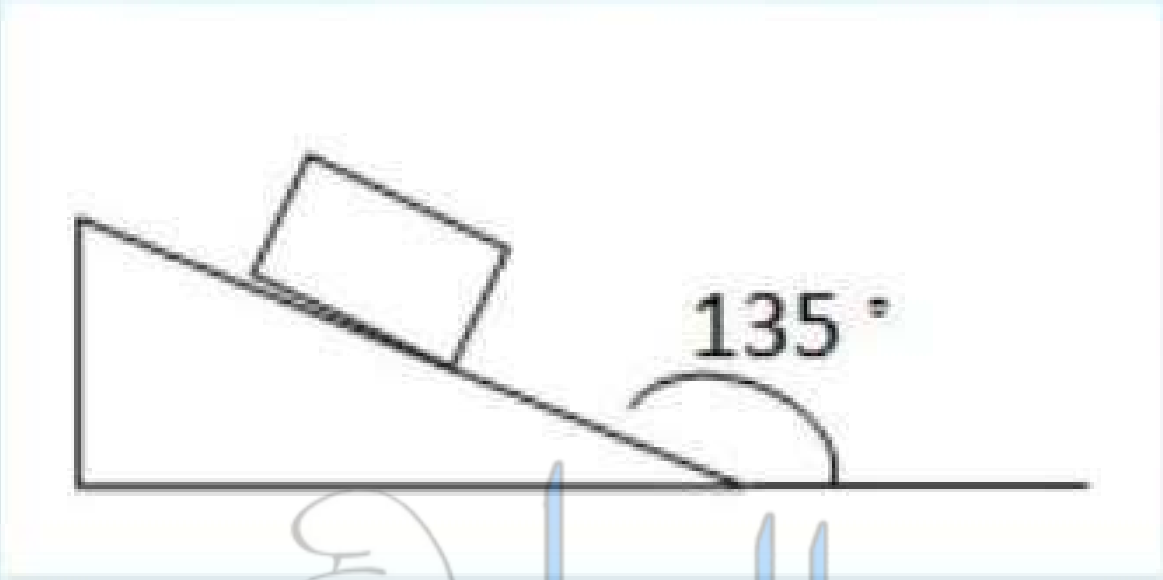
If pendulum take time (0.067s) to move from (a) to (C), then the period of one oscillation (T) :



Select one:

- A. 0.134 S
- B. 0.67 S
- C. 0.067 S
- D. 0.268 S

(m) is about to slide 0ver the inclined level shown. so (μ_s):



Select one:

- A. 1
- B. 0.6
- C. -1
- D. 0.4

In a gravitational potential energy experiment, when (m) reaches Earth, the mass (M)

Select one:

- a. **The speed of the mass (M) will decrease**
- b. **The mass(M) will stop.**
- c. **The speed of the mass (M) will increase.**
- d. **The mass (M) will move in costant speed.**

The relation between the mass of pendulum (m), and the periodic time (one osiliasion time) (T) at the same lenght of pendulum thread



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A. inverse linear

B. linear direct

C. curve relationship

D. no relation

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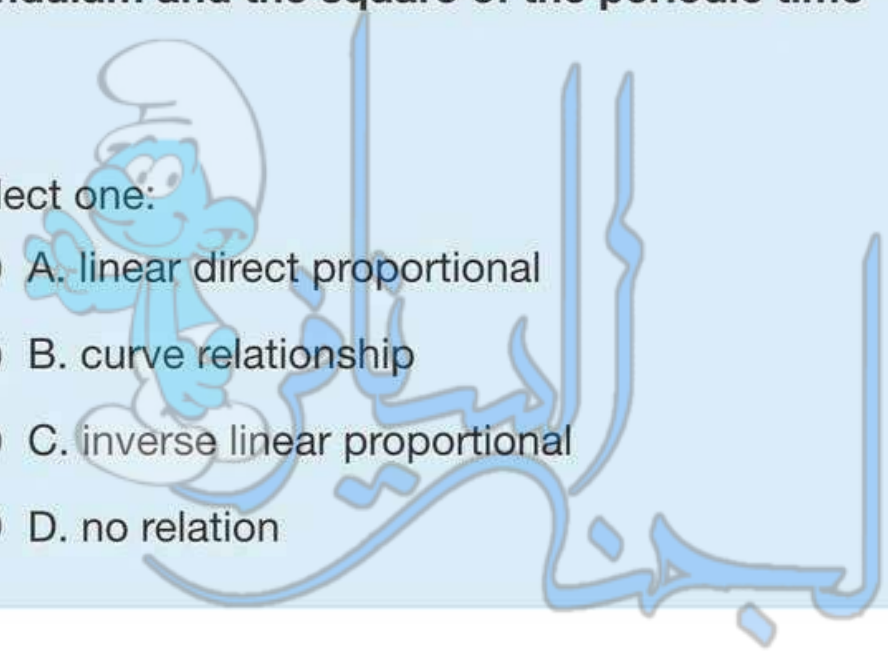
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question

The relationship between the length of the thread of the pendulum and the square of the periodic time

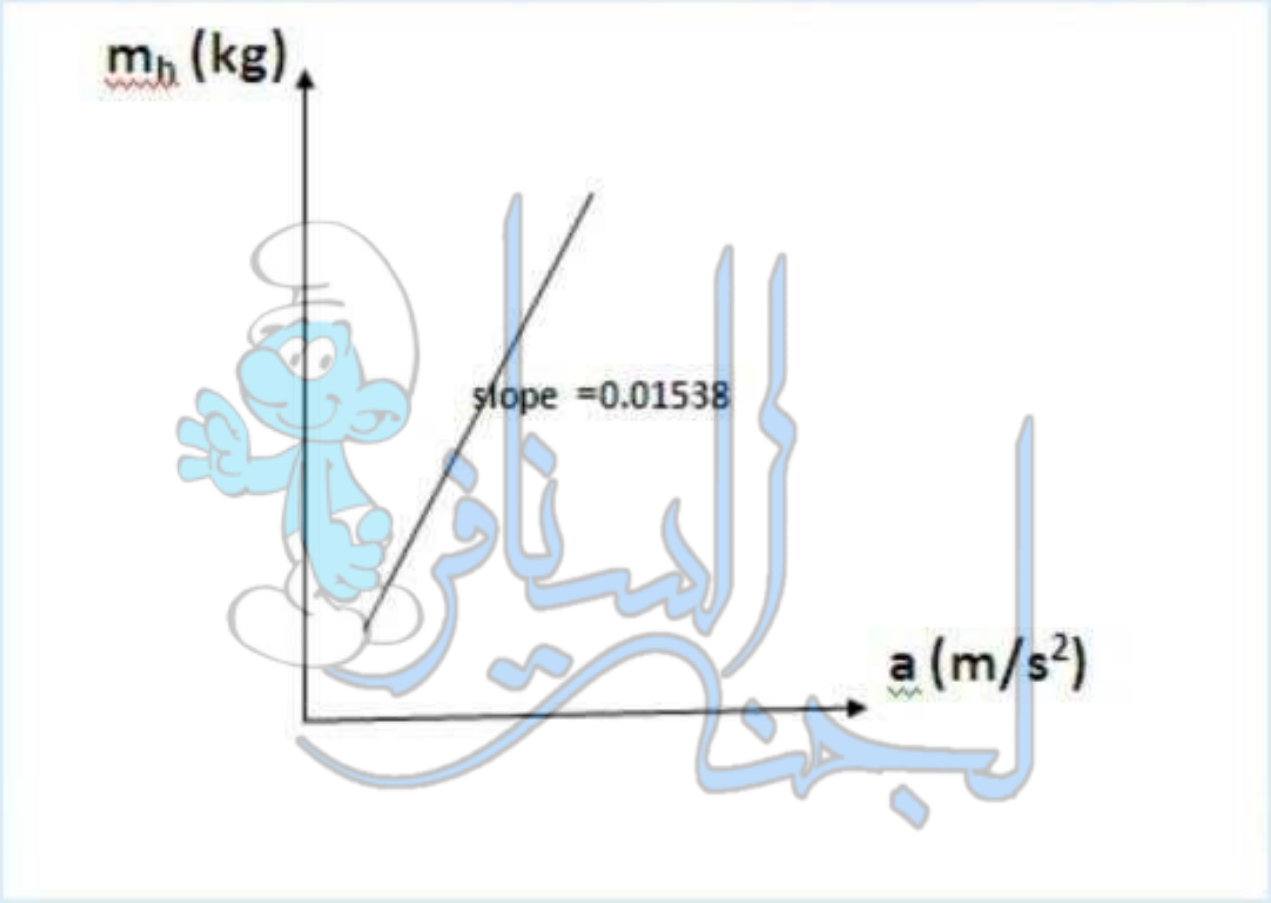
Select one:

- A. linear direct proportional
- B. curve relationship
- C. inverse linear proportional
- D. no relation



Next page

In Newton's second law experiment, a student obtained the graph shown, when the total mass was .(constant(200g
Depending on his drawing, the magnitude of the acceleration of the Earth's gravitational pull



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a. **12 m/s^2**

b. **11 m/s^2**

c. **9 m/s^2**

d. **13 m/s^2**

Suspend a body of mass (30 g) with a spring, if you know that the periodic time for the existence of this mass and when the elongation is fixed is 0.37 s.

The spring constant (K) is:

اختر أحد الخيارات

a. 8.6 N/m

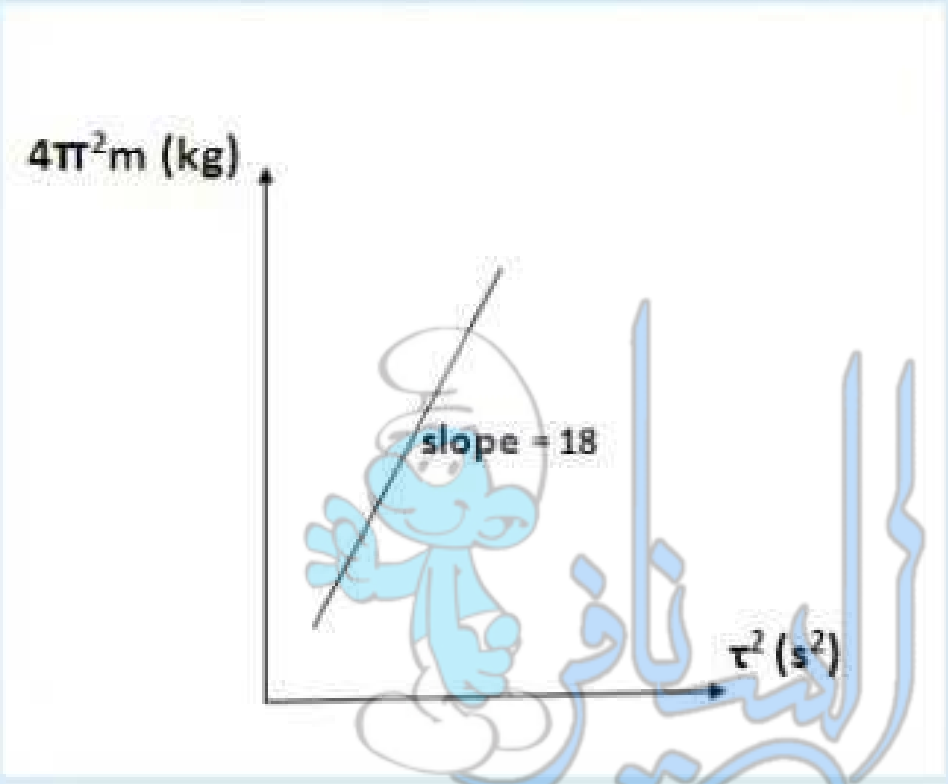
b. 6 N/m

c. 12 N/m

d. 10 N/m



In Hook's law experiment, student draw the relation shown , the spring constant (k):



Select one:

- A. 10 N/m
- B. 3 N/m
- C. 18 N/m
- D. 9 N/m

The unit of spring constant is (kg/s^2).

Select one:

- True
- False

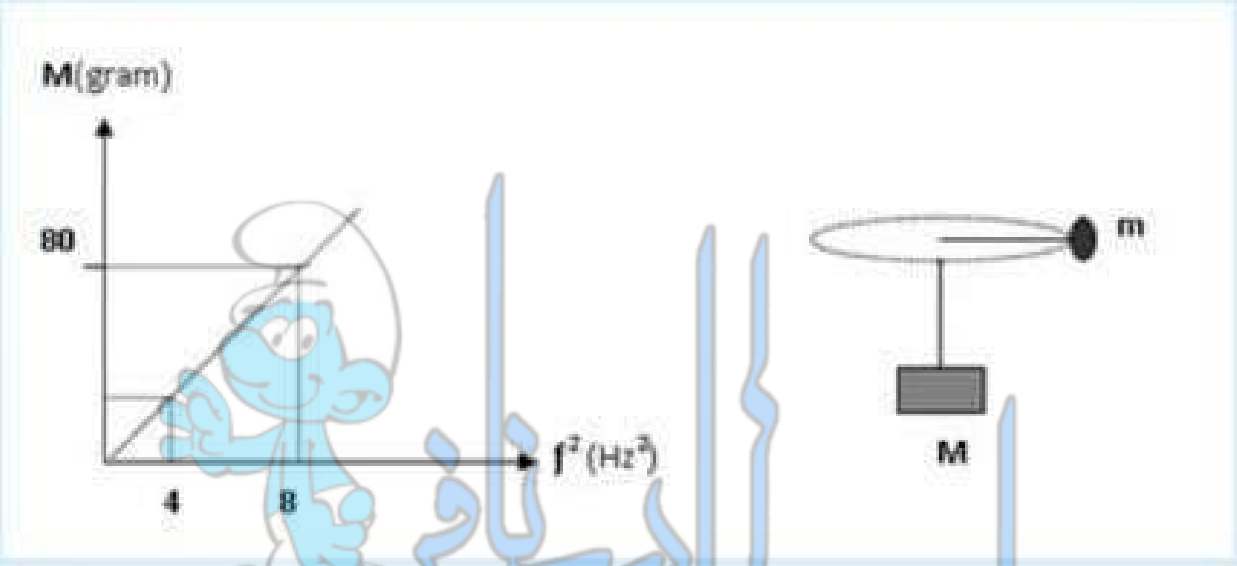


In a gravitational potential energy experiment, when (m) reaches Earth, the mass (M)

Select one:

- a. **The speed of the mass (M) will decrease**
- b. **The mass(M) will stop.**
- c. **The speed of the mass (M) will increase.**
- d. **The mass (M) will move in costant speed.**

In the circular motion experiment, a student studied the relation between the hanging mass (M) and the square of the frequency (f^2) for rotational body (m) using the radius (R=0.3 m). A plotted data are shown in the figure bellow. The mass (m) of the rotational body is:



Select one:

- a. Non of above
- b. 8.45 gram
- c. 4.22 gram
- d. 6.34 gram
- e. 12.7 gram

In a gravitational potential energy experiment, if the initial height of the mass (m) is (h_i = 1 m)and its mass (20g).

The final kinitec energy ($(K.E)_f$, when the mass (m) reachs the height (h_f = 80cm) is:



Select one:

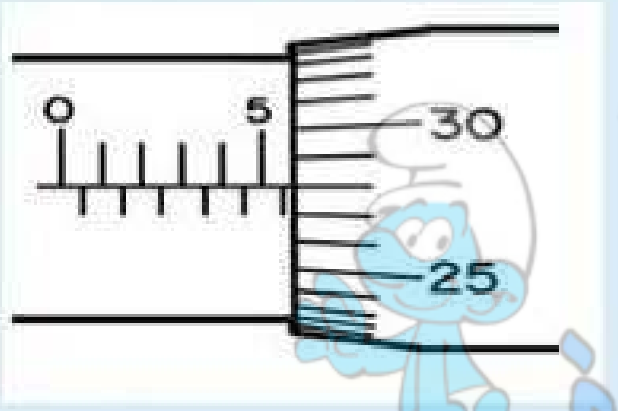
- a. **0.04 J**
- b. **2 J**
- c. **0.3 J**
- d. **0.4 J**

Square-shaped plate with side length (L).The uncertanity (error) in it area is:

Select one:

- a. $4L.\Delta L$
- b. $4L^2.\Delta L$
- c. $2L^2.\Delta L$
- d. $2L.\Delta L$

In an experiment to measure the density of a small metallic sphere, the diameter of the sphere (d) was measured by micrometer shown in the figure, and the mass of the sphere M was $M \pm \Delta M = (2.1 \pm 0.05)$ g. calculate the density of the sphere, density ± error density



Select one:

- a. 20792 ± 602.97 ✓
- b. 10762 ± 302.96
- c. 30552 ± 56.77
- d. Non of above
- e. 24893 ± 652.98

A cube with length (L), the error in the surface area:



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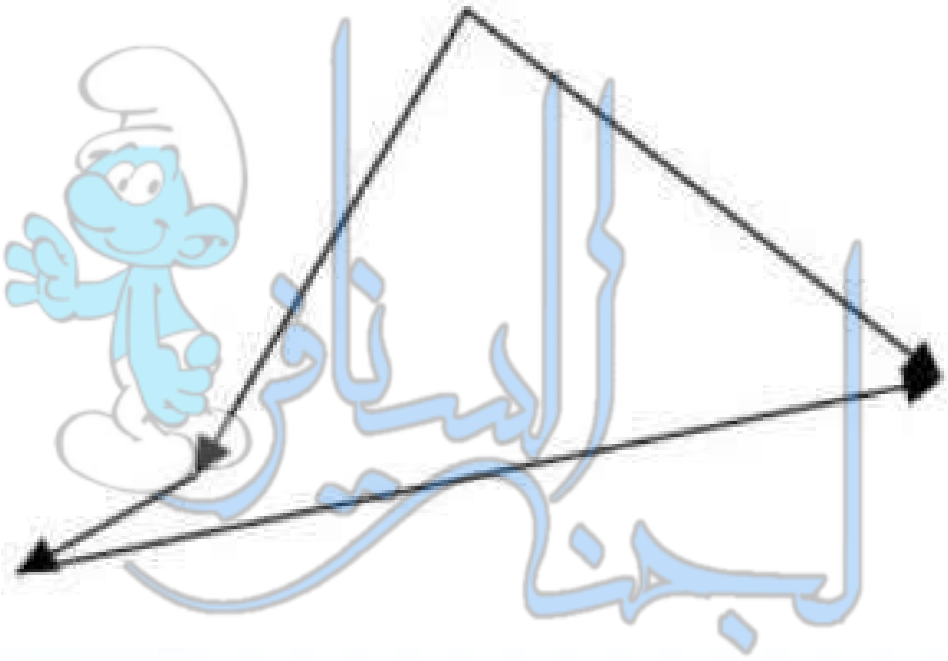
A. $12\Delta L.L$

B. $6\Delta L.L$

C. $12\Delta L.L^2$

D. $6\Delta L.L^2$

In the shown vector addition, the resultant of forces is zero



Select one:

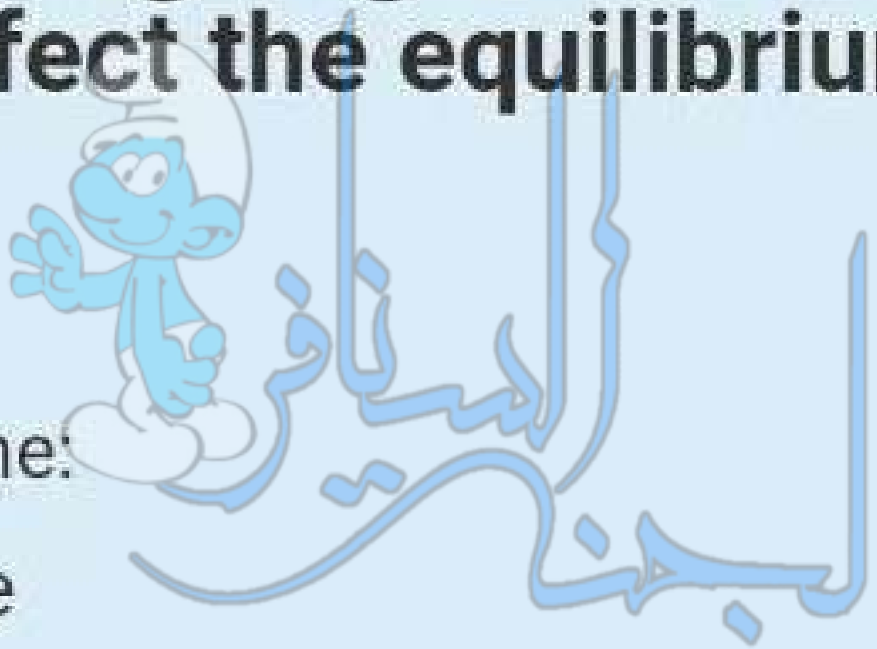
- True
- False

In a balanced system on the force table, adding 30g to each strapless will remain balanced

Select one:

- True
- False

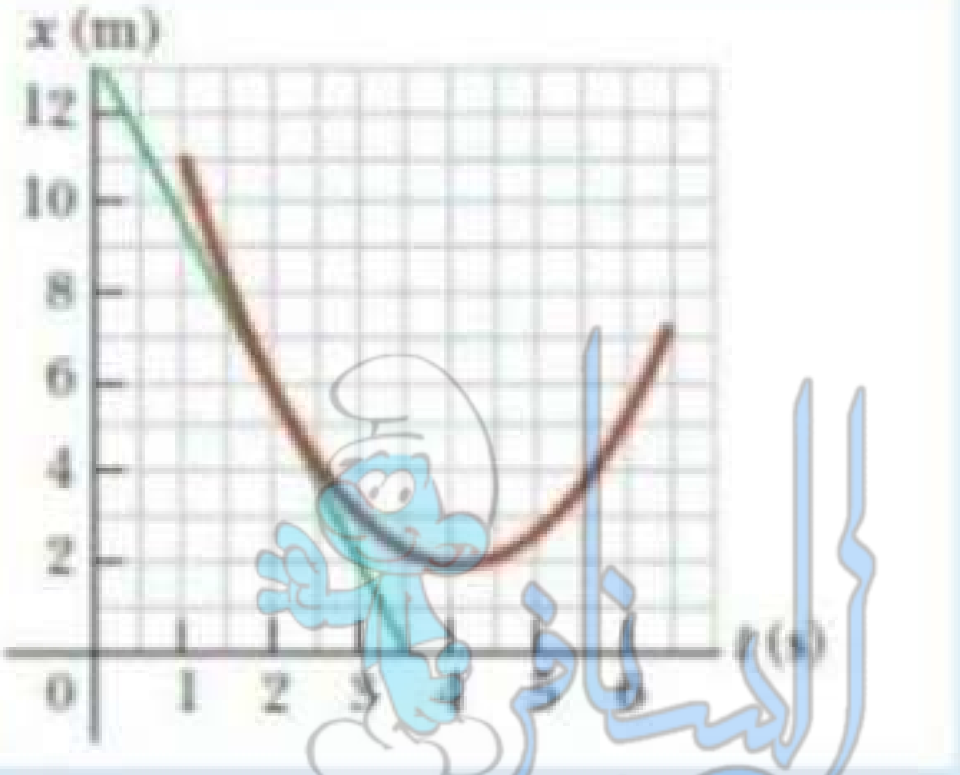
In a vector experiment, the ring being large or small does not affect the equilibrium



Select one:

- True
- False

A position-time graph for a particle moving along the x-axis is shown in the next figure. What is the velocity at t= 2 s?



Select one:

- a. -3.8 m/s
- b. 3.71 m/s
- c. Cannot be determined from the graph
- d. 3.8 m/s
- e. -3.71m/s

A balanced system of three forces, if the sum of (F2) and (F3) equals

then the magnitude of (F1) ,(0.89N)

((0.89)N

إختر واحداً:

- صح
- خطأ

A cube with length (L), the error in the surface area:

إختر أحد الخيارات:

A. $12\Delta L.L$

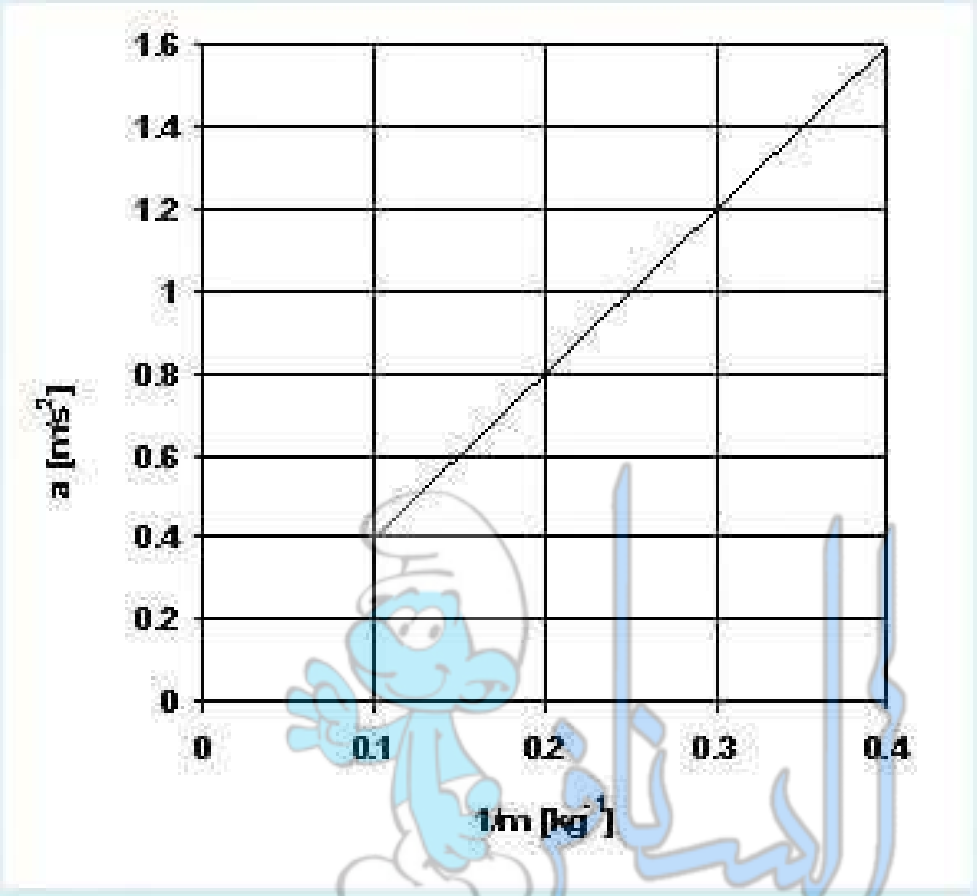
B. $6\Delta L.L$

C. $12\Delta L.L^2$

D. $6\Delta L.L^2$



In the Newton's second law experiment, the relation between the acceleration (a) and the total mass (m) with a constant force (F) is shown in Figure. The applied force on the system (in N) is:



Select one:

- a. 12 N
- b. 8 N
- c. Non of above
- d. 16 N
- e. 4 N ✓

The correct answer is: 4 N

If F is the force, x the displacement, and k a spring constant, for simple harmonic motion, F is given by:

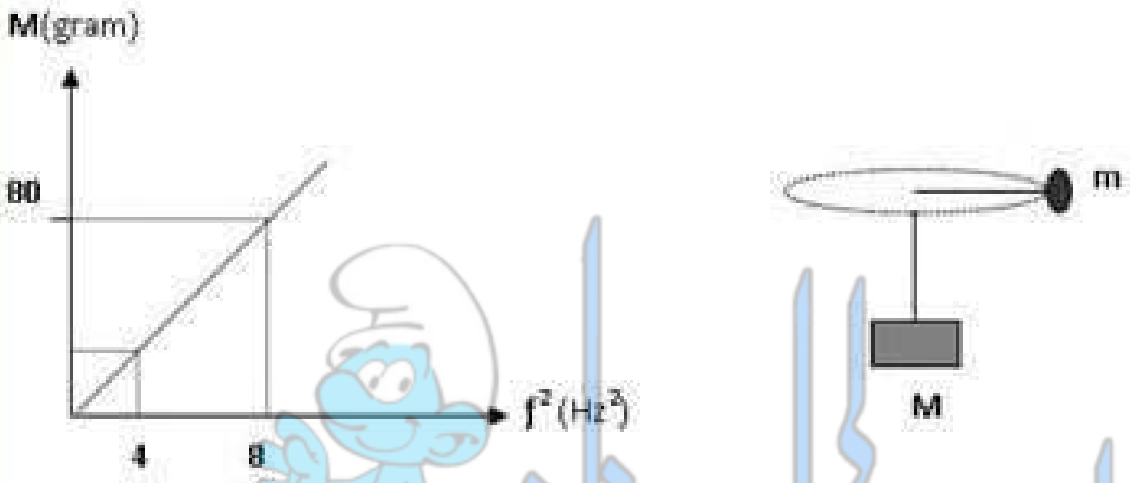
Select one:

- a. F=kx
- b. F = -kx
- c. F=-k/x ✗
- d. $F = -kx^2$
- e. Non of above



The correct answer is: F = -kx

In the circular motion experiment, a student studied the relation between the hanging mass (M) and the square of the frequency (f^2) for rotational body (m) using the radius (R=0.3 m). A plotted data are shown in the figure bellow. The mass (m) of the rotational body is:

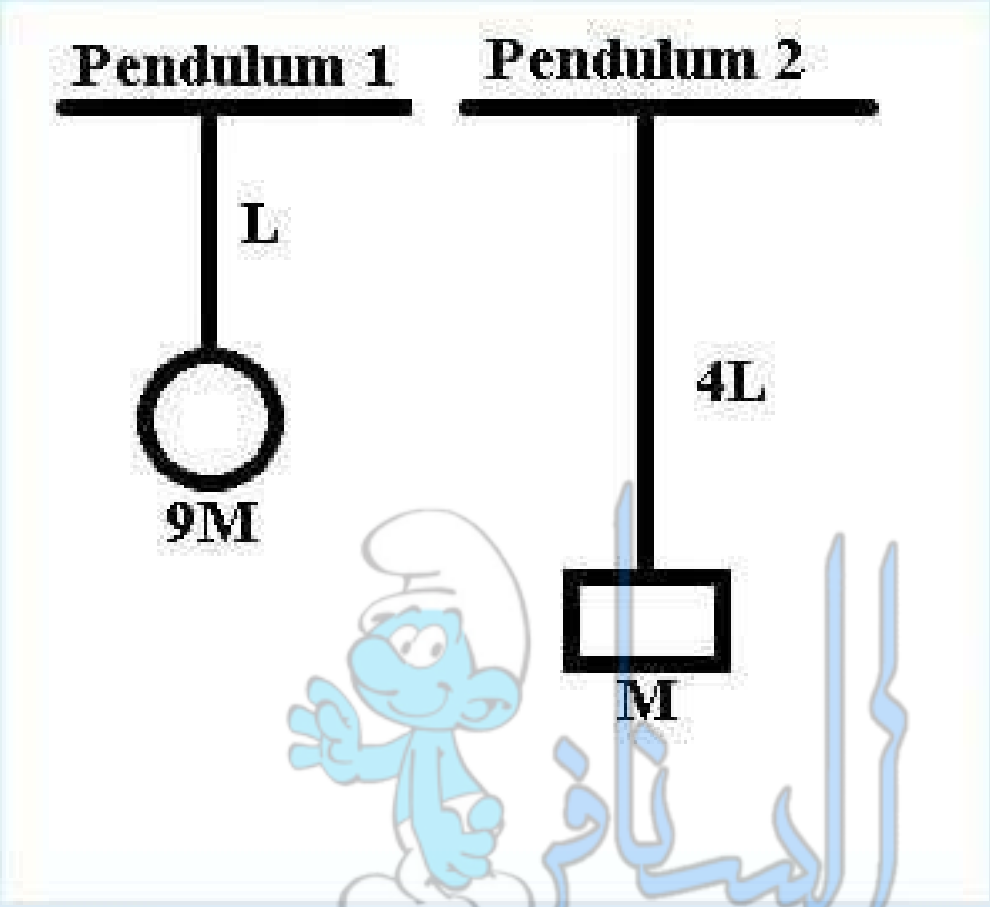


Select one:

- a. 6.34 gram
- b. 8.44 gram ✓
- c. 12.7 gram
- d. Non of above
- e. 4.22 gram

The correct answer is: 8.44 gram

In the simple pendulum experiment, the following figures show two pendulums. If the periodic time is T_1 for the Pendulum (1), and the periodic time is T_2 for the pendulum (2), then the relation between T_1 and T_2 is



Select one:

- a. T2 = 2.45T1
- b. Non of above
- c. T2 = 2.24T1
- d. T2=2T1 ✓
- e. T2=1.73T1

The correct answer is: T2=2T1

Micrometer reading is (8.74 mm), if we turned it (8) turns and (3/5) of turn, towards opening direction, its reading will be:

Select one:

- A. 5.34 mm
- B. 13.04 mm
- C. 4.54 mm
- D. 12.14 mm

A student used flexible ruler with maximum range (30 cm) to find circumference of circular table. If the radius of the table (80cm), then the uncertainty in circumference (ΔC) is:

إختر أحد الخيارات:

- A. 17 mm
- B. 8.5 mm
- C. 10.5 mm
- D. 5.5 mm

In the simple harmonic oscillator, a mass of (40g) is hanged in a spring, if the extension of the spring is (4 cm) and if the mass drawn to (3cm) from equilibrium point and oscillated 25 times, the total time of an oscillation of 20 times is (in sec):

Select one:

- a. 9.93
- b. 11.91
- c. Non of above
- d. 5.96
- e. 7.95 ✓



The correct answer is: 7.95

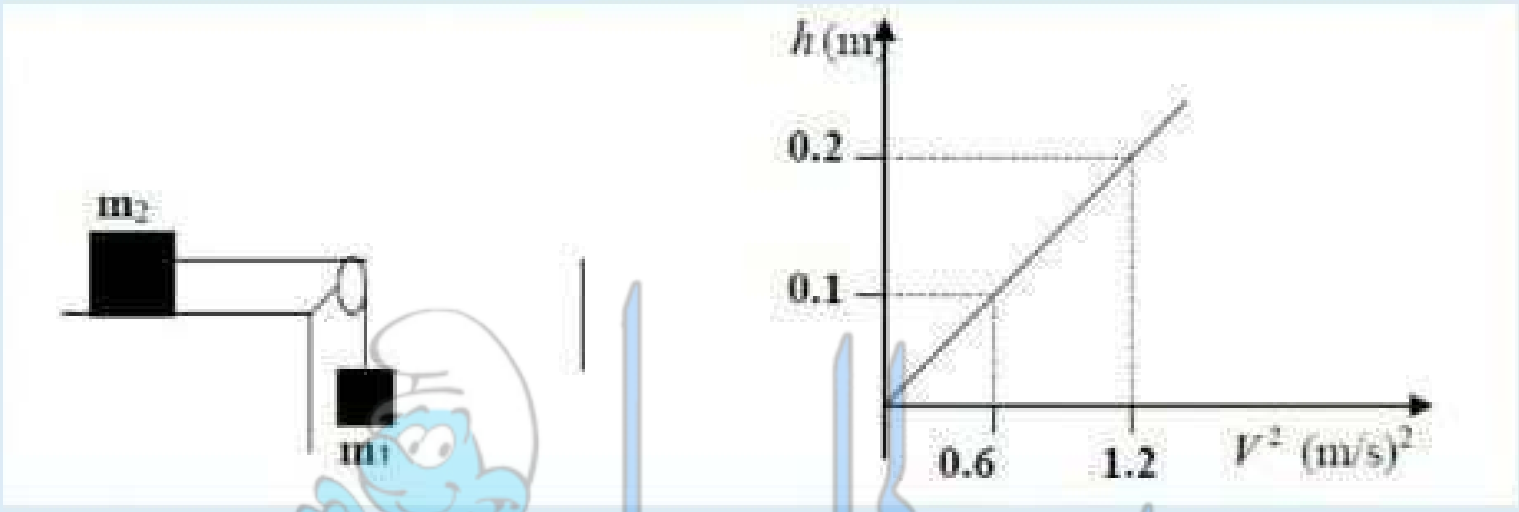
In a vector experiment, the ring being large or small does not affect the equilibrium



Select one:

- True
- False

In the conservation of energy experiment, a student obtained data plotted in the following figure. If the masses m_1=0.15 kg and m_2=0.2 kg, the acceleration of gravity (in m/s^2) is:

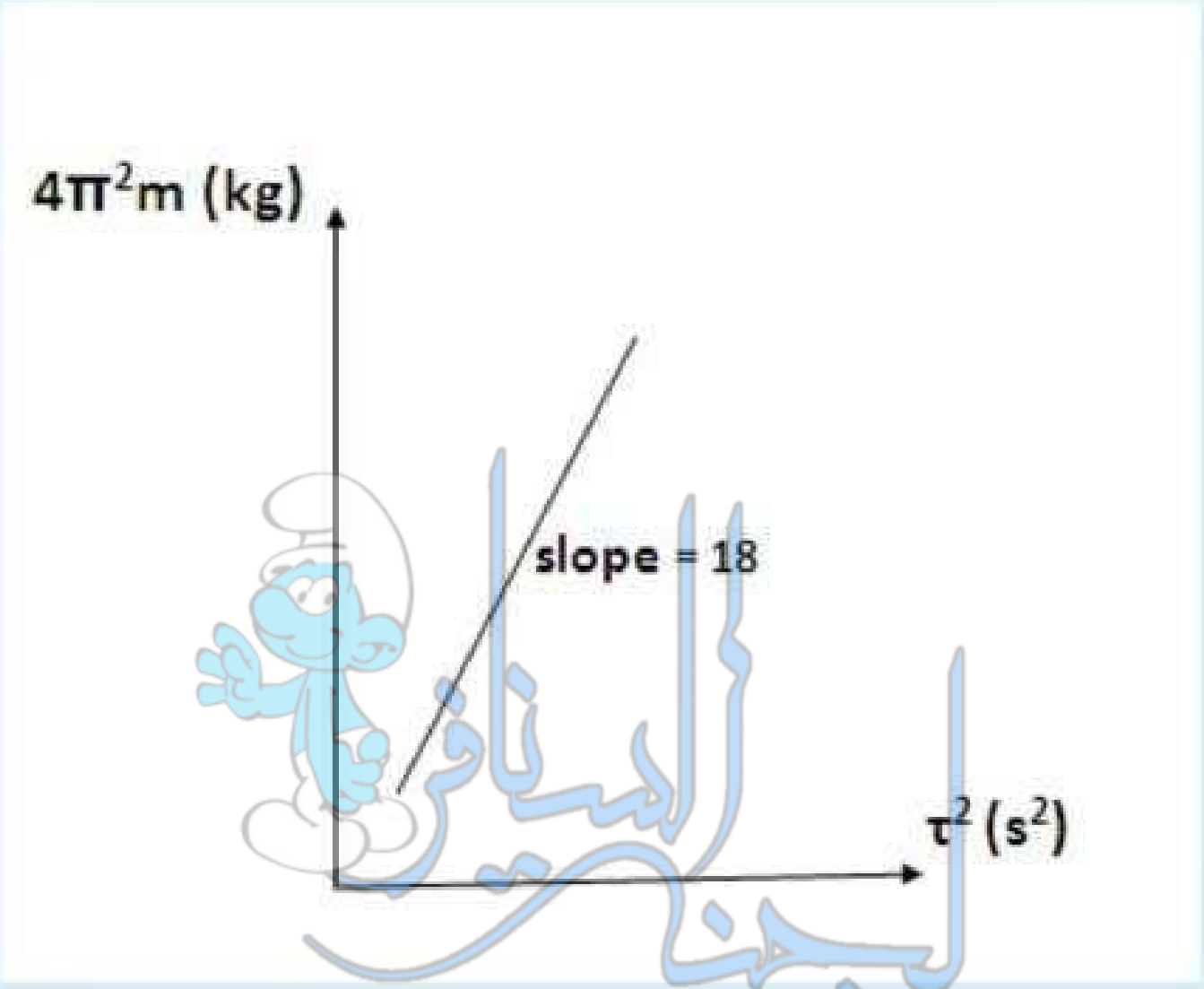


Select one:

- a. Non of above
- b. 11
- c. 7 ✓
- d. 9
- e. 13

The correct answer is: 7

In Hook's law experiment, student draw the relation shown , the spring constant (k):



Select one:

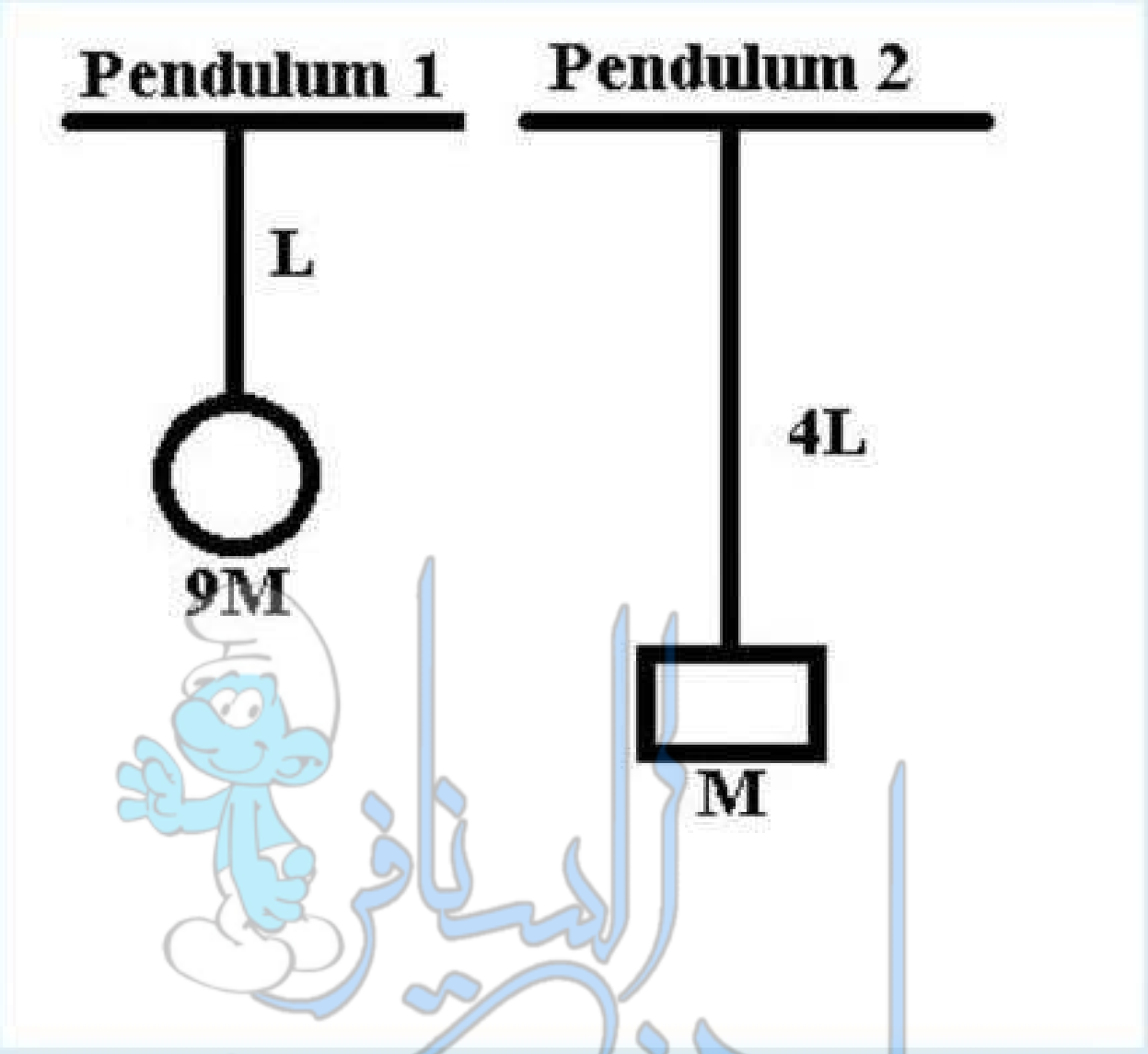
- A. 3 N/m
- B. 18 N/m
- C. 10 N/m
- D. 9 N/m

we can say about (μs) and (μk) that :

Select one:

- A. **-1≥ (μs)AND (μk) ≥ 1**
- B. **(μs) ≥ (μk)**
- C. **(μs)< (μk)**
- D. **(μs)AND (μk) ≥ 1**

The two pendulums shown have the same :



Select one:

- a. **acceleration of gravity**
- b. **predic time**
- c. **angular frecuancy**
- d. **all of them**
- e. Non of above